

Bangladesh University of Engineering and Technology

Department of Industrial and Production Engineering



IPE 304

Product Design I Sessional

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FORWARDING LETTER

Date: 10/09/23

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Subject: Report on “Development of a Smart Pop-Up Table.”

Honorable teachers,

With due respect and humble submission, we would like to submit the report on ‘Smart Pop-Up Table’. Currently available conventional table lacks flexibility and is unfunctional. For this, we have decided to develop a table which utilize the space as much as we can and to give flexibility to the users. The operating concept of the ‘Smart Pop-Up Table’, functional decomposition, manufacturing process, and total cost of production are all shown in this report.

We would like to express our heartiest gratitude for helping us with the recommendations for carrying out this project. There might be some error or lacking as this is first functional prototype of the product. We, pray and hope that you will be kind enough to consider such cases and oblige thereby.

Sincerely yours,

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PREFACE

Engineering design is a systematic process through which an engineer creates functioning goods and processes. It is an iterative process. Often, portions of the procedure must be performed several times. We have studied how to develop a product and do the necessary analyses to build it in Product Design-I Session. The purpose of this project was to put the information obtained during the Product Design Session into practice. After talking with our course teachers, we were assigned course project " Smart Pop-Up Table." Currently, available conventional table is lacks flexibility and unfunctional. However, with our suggested solution, we provide a table to the user which is more flexible and multifunctional.

We began our report by conducting a survey of our product among students, mess and sublet residents, small organization, and others. After that, we conducted a QFD for our product. Following that, a qualitative examination of material selection, manufacturing process selection, and joining process selection was performed. After that, we selected the appropriate material, manufacturing techniques, and joining procedure for each component of our product utilizing digital logic and performance indices. After completing the design study, we conducted a cost analysis to determine all expenses associated with our product.

Hopefully, this report will encourage students to creative designing of a product.

SUMMARY

Everybody aspires to make their everyday tasks easier than they were previously thanks to technology innovation. With its great features, Smart Pop-Up Table will make life much simpler for those who must spend a lot of time at a desk for work, such as students. Our aim is to design a new modified table which is ergonomic featured, more flexible. Here we try to gather data from field to realize the market and try to make a feasible solution. We use the data finding the most important requirement for customers or producers. Then, we find proper material for each part of our machine and check feasibility, either it is bought or manufactured. Then we make a design using 'SolidWorks' and analyze different parts using 'Ansys'. No business can run without a financial plan. So, finally, we make a financial plan, assign cost for different scenario and find out the breakeven. Also, we test the sensitivity by changing different parameters. We hope this report will be an asset for developing smart tables in the future.

ACKNOWLEDGEMENT

At first, we thank the Almighty for His graceful sight upon us.

Secondly, with due respect, we would like to thank our honorable faculty Dr. Ferdous Sarwar, S. M Shabab, Kaniz Fatema for their patience, guidance and support in the preparation of this report and this course. It feels great to have been given an opportunity to do a real-life work related to execute new product idea and related skill development. We will always be indebted to him for the valuable suggestion and all the time that he has spent in guiding us through the course. Without his immense support this project would not have been possible to complete at all, least of all, within given time.

Lastly this report could not be attributed with a single person's effort only. Each member of the group contributed equally to the completion of this report. Every team member contributed an equally important part putting this report together. Again, we would like to thank our honorable faculty and everyone who helped us by cooperating with us.

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Chapter 1: Introduction

1.1 Product Design

Product design is the process of conceptualizing and ultimately materializing these concepts into tangible inventions, by combining art, science, technology etc. This begins with problem definition and idea creation. A team of designers seek to better understand the issues users face, and then come up with a feasible solution that will solve the customers' needs within the constraints laid out. Feasibility includes determining the cost of the product, the potential problems it might face & how difficult it is to manufacture. A prototype is thus developed. Finally, the prototype enters testing, where the designers finalize the product through continuous improvements, and then the product is ready for marketing. Product design requires a deep understanding of the user. The better the designers understand their customers, the more effective and relevant the product design outcome will be.

1.2 Proposed Product Ideas

In the beginning, we have selected these six designs which included:

1.2.1 The “Multi Cultivator”

It is a combination of insecticide and fertilizer sprayer. Multiple Sprayers will be attached to the front of a movable cart. Electrically controllable Seeder cum fertilizer equipment will be added at the bottom of the cart. Manually adjustable plough will be added alongside handle of the cart which can be used as per need. It is multifunctional and efficient. It will lower aggregate cost of cultivation.

1.2.2 Multipurpose automated sieving machine

It is an automated sieving machine which will fetch the material from stack and filter them automatically. The sieve will be attached to a movable frame by bearings. A belt drive mechanism will connect the sieve to an electric motor so that the sieve moves back and forth continuously. Conveyor belt will be added to frame. Wheels will be attached at the bottom of the frame to make it portable. Sieve of different sizes can be used in a single machine. It can be used in agricultural, construction sectors and in many cases where sieving is done manually.

1.2.3 Garden Cleaner Pro

Leaf is sucked and ground into powder. This makes it possible to collect huge number of leaves in one attempt. Leaf is sucked through suction using vacuum pressure. Sieve is placed inside the machine which hinders rocks or soil particles from entering the grinder. Automatic grinder is accommodated in the pathway. A vessel is attached for collecting the ground leaves. Ground leaves easily get decomposed compared to whole leaves; thus, are great as fertilizers.

1.2.4 Fold-able Forklift Mini

It is a smaller version of forklift which is a combination of product lifting mechanism and folding mechanism. With fold-able mechanism, it can be used for shipment through narrow passages. It can be used in small scale operations too. It is easy to carry and comparatively cheaper.

1.2.5 Multi-Purpose Dumpling Machine

Using this single machine, we can make different kinds of foods of different dough and fillings like shingara, samosa, momo etc. It will have separate containers for dough and filling. This machine will not only give different shapes as per need but will also fry the food. It will give us ready to eat food. Using it, customers will be served hot and fresh food instantly. It will save time and increase productivity.

1.2.6 Smart Pop-up Table

The main purposes of the smart pop-up table are to utilize the space as much as we can and to give flexibility to the users. This table will have smart computer-controlled lock system as well. Besides, the table will be portable, ergonomic and will have additional beneficial features.

1.3 Selected Product

After having proper discussion with our course teachers, the last proposal “*Smart Pop-Up Table*” was selected as our project.

1.3.1 Background

Most of the students in colleges and universities pursuing higher education or about to pursue higher education live in messes or sublets sharing a single room.

Those who live in messes, sublets or dormitories need to share a single room with two or three of their fellows. If we look at our varsity dormitories, we can see how our friends are sharing rooms with 3 others. They lack proper space management. Their rooms look messy. Things are scattered here and there. The room is further cluttered with wires from chargers and multi-plugs.

It is not possible to solve this problem by buying new products as those will only occupy more space. The only way to solve this problem is to make all these conveniences available in a single product that is a part and parcel of our daily life. It will also make good use of the space.

In quest of solution to all these problems, we think “Smart Pop-Up Table” is one step advancement towards a feasible change to remove the problems.

1.3.2 Main Features

- App Controlled Safety Lock: Our product will have App-controlled safety mechanism. Its shelves will have app-controlled lock system.
- Pop Up Mechanism: Computer controlled pop up mechanism will give more flexibility to the users. Its shelves, cabinet and drawers will be brought up only while in use or when it is needed. Thus, our smart table will serve as both a flat table as well as a study table which will assure efficient space utilization.
- Ergonomic: We will allow the users to adjust the table’s height according to user’s position and posture to give smooth user experience. Moreover, the computer monitor’s height can be adjusted to reduce strain on the nape.
- Power Socket: Our table will have built-in power socket to recharge laptop or mobile phone.
- Orange Pi PC: The smart table will have built-in Orange Pi PC with 17-inch display attached to perform browsing or official tasks.

1.3.3 Improvements

Conventional study table lacks flexibility. The ergonomic features that we have introduced with our smart table will give more flexibility while in use, as the motion sensor will observe the user’s motion and adjust the table’s height as well as the monitor’s height. Conventional study tables have safety issues also. It is very troublesome to keep different keys for different shelves, drawers and cabinets and also other safety issues. Using our smart table, one will be able to lock drawers and cabinets using a single computer only. The table will also have rechargeable power storage so that user can use laptop or recharge smartphone during load shedding. Our smart table will have power sockets of its own to use and recharge devices. It will have wheels with its legs so that it can be shifted easily.

Conventional study table is unfunctional but our pop-up mechanism will enable the table to be multifunctional. It will be working as both flat table and study table. Hence space will be utilized efficiently. It will have in built computer with 14-inch display that will help one to browse or do official tasks.

Chapter 2: Understanding Customer Needs Through Survey

2.1 Introduction

A study table is essential for everyone, from children to the elderly. We can use the study table for a variety of things, including reading, writing, and using laptops, computers, and storage. A study table is a spot where you sit every day bringing consistency and discipline to your life. The fix amount of area that provided by the normal table couldn't give the user the flexibility to use it. Sometimes a student might need to use a computer. Students keep their necessary documents in the drawer of their study table. If a digital locking system is introduced, it can enhance the security. Also, a student might don't want the monitor always on the table. To deal with all these situations Smart Pop-Up Table is introduced.

With technological advancement everybody wants to make their daily works easier than before. Smart Pop-Up Table will help the people who have to spend a lot of their time on desk for their work like students to make their life much easier with its wonderful features.

As primary research, we identified our main customers are:

- Students
- Mess and sublet residents

We carried out a survey for developing customer requirements for our Smart Pop-Up Table among some of the above customers in a very short period of time.

2.2 Customers and Locations of Survey

2.2.1 Locations of Survey

1. BUET dormitory
2. Sublets around BUET
3. KUET dormitory
4. RUET dormitory

2.2.2 Customers

1. Students
2. Mess and Sublet residents
3. Small Organization

Table 1: Response Table

Location	No. of response
BUET Dormitory	27
Sublets around BUET	13
KUET Dormitory	10
RUET Dormitory	9

2.3 Survey Results

For better understanding our customers, we conducted a survey to find out what situation our targeted customers are in, what are the features that they are looking for in a smart study table and their expected price and service life of the product. %9 people participated in this survey. Using the result of this survey, we created pie charts and bar charts in order to visually comprehend their needs.

The results of the survey are depicted below in tables as well as graphically using pie charts or bar charts:

1. Have you heard of smart furniture?

- Yes
- No

Table 2: Response count of respondents' knowledge of smart furniture

Options	Response Count	Response Percentage (%)
Yes	27	45.8
No	32	54.2

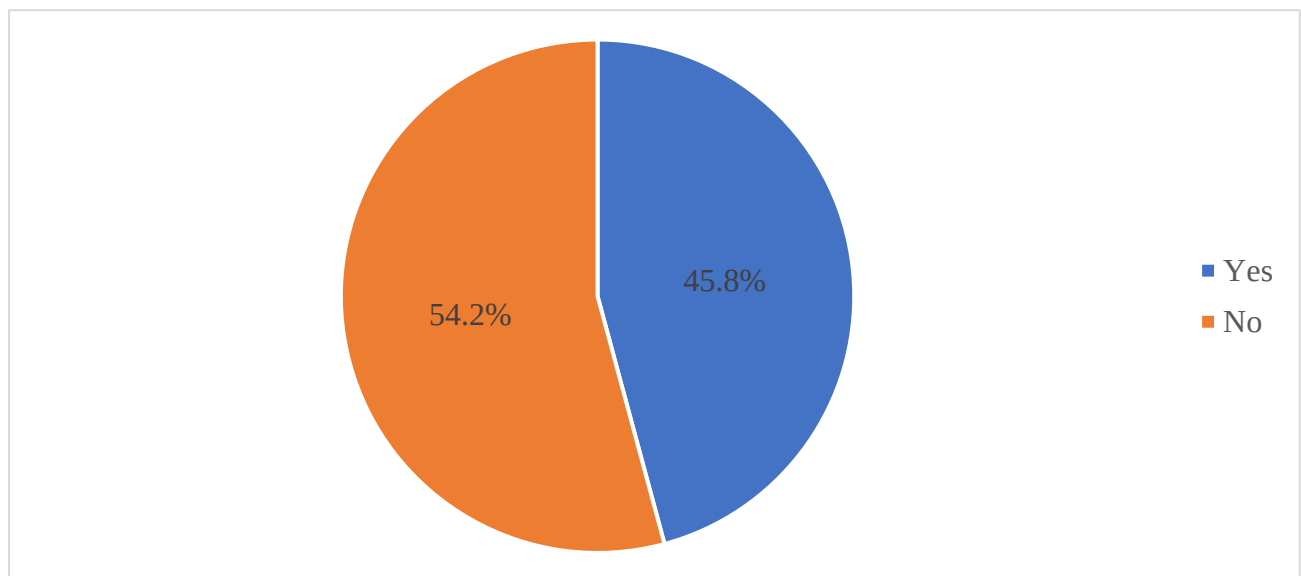


Figure 1: Pie chart of respondents' knowledge of smart furniture

2. Do you feel the need for automated mechanism in your study table that will make your life much easier?

- Yes
- No

Table 3: Response count of respondents' want of automated mechanism in their study table

Options	Response Count	Response Percentage (%)
Yes	48	81.4
No	11	18.6

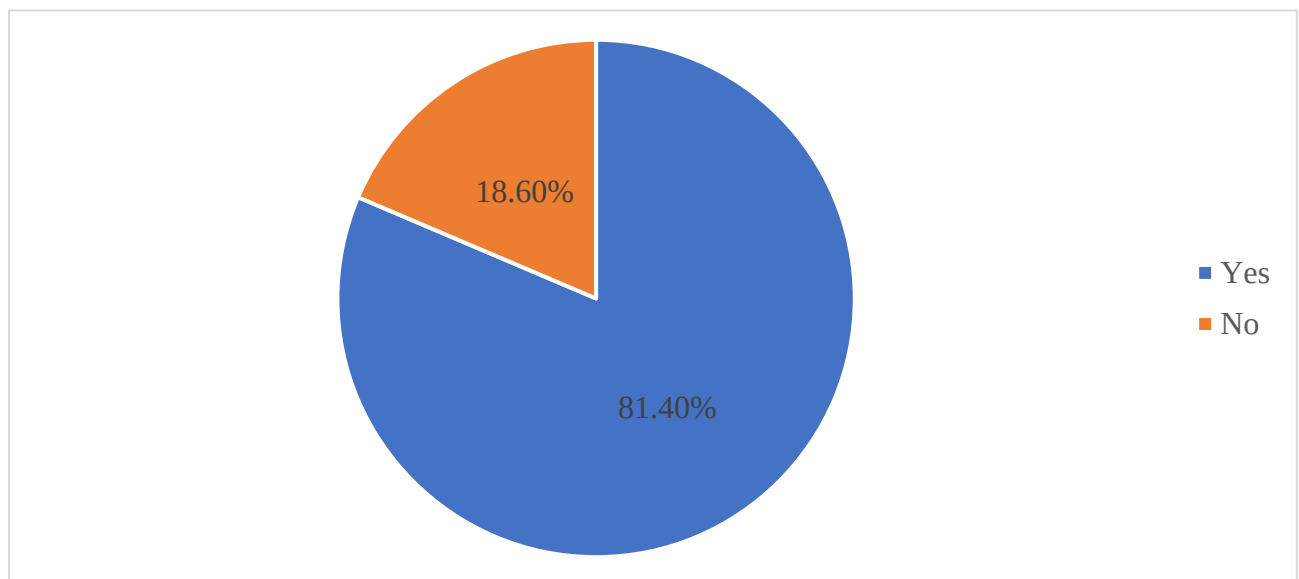


Figure 2: Response count of respondents' want of automated mechanism in their study table

3. What do you think about the market demand for smart furniture?

- Low
- Medium
- High

Table 4: Response count of respondents' knowledge of market demand for smart furniture

Options	Response Count	Response Percentage (%)
Low	13	22
Medium	32	54.2
High	14	23.7

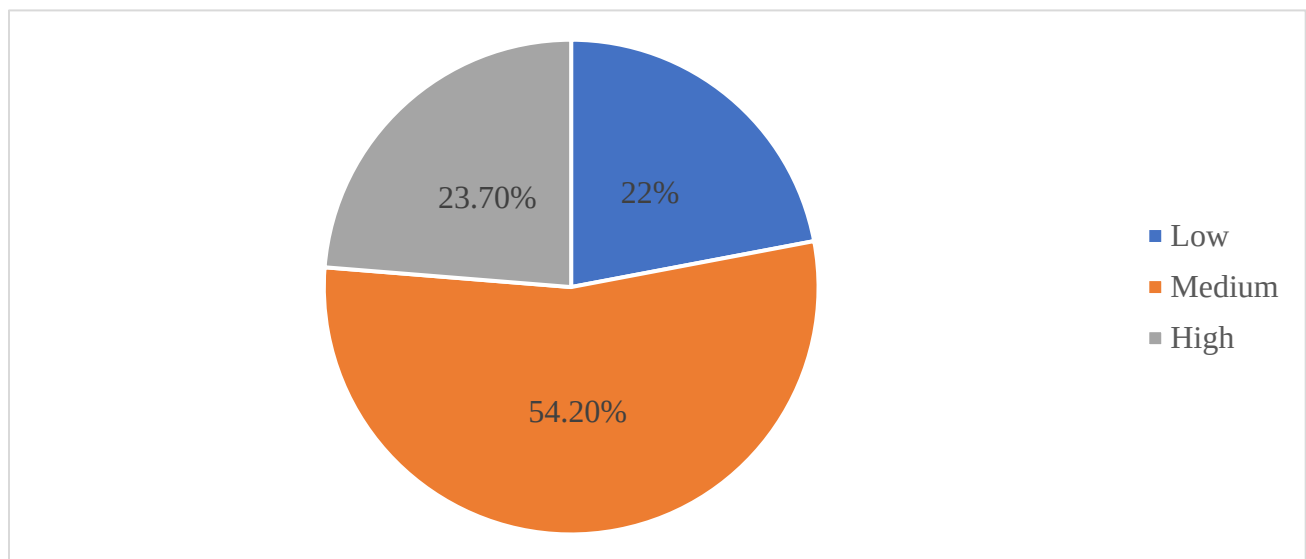


Figure 3: Pie chart of respondents' knowledge of market demand for smart furniture

4. Where do you live for your professional purpose?

- Family Apartment
- Dormitory/Hall
- Mess
- Sublet

Table 5: Response count of respondents' place of living for professional purpose

Options	Response Count	Response Percentage (%)
Family Apartment	25	42.4
Dormitory/Hall	17	28.8
Mess	14	23.7
Sublet	3	5.1

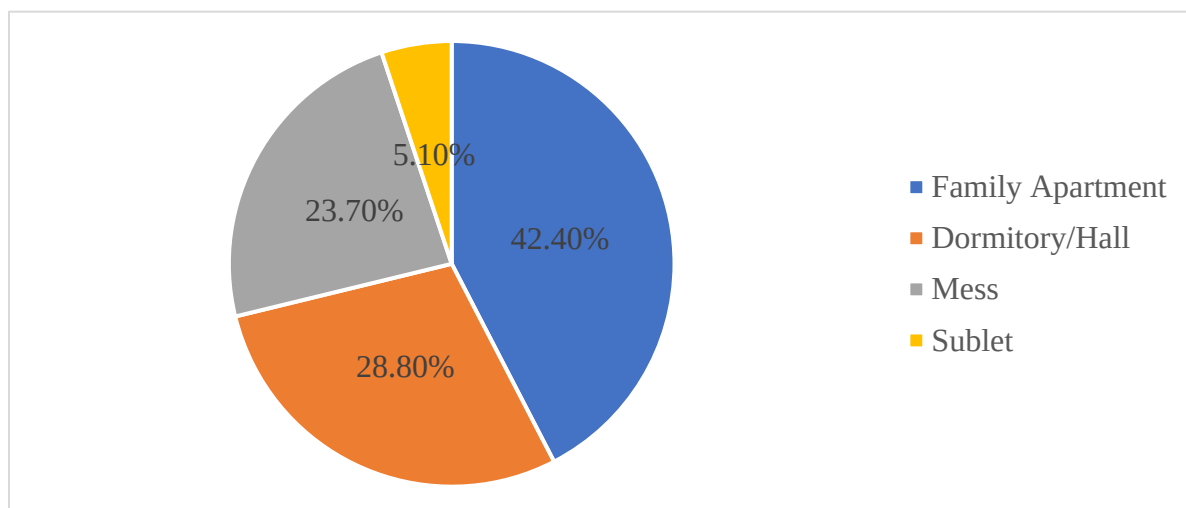


Figure 4: Pie chart of respondents' place of living for professional purpose

5. What is your current study table made of?

- Wood
- Metal
- Wood and Metal
- Other

Table 6: : Response count of respondents' study table material

Options	Response Count	Response Percentage (%)
Wood	40	67.8
Metal	1	1.7
Wood and Metal	10	16.9
Other	8	13.6

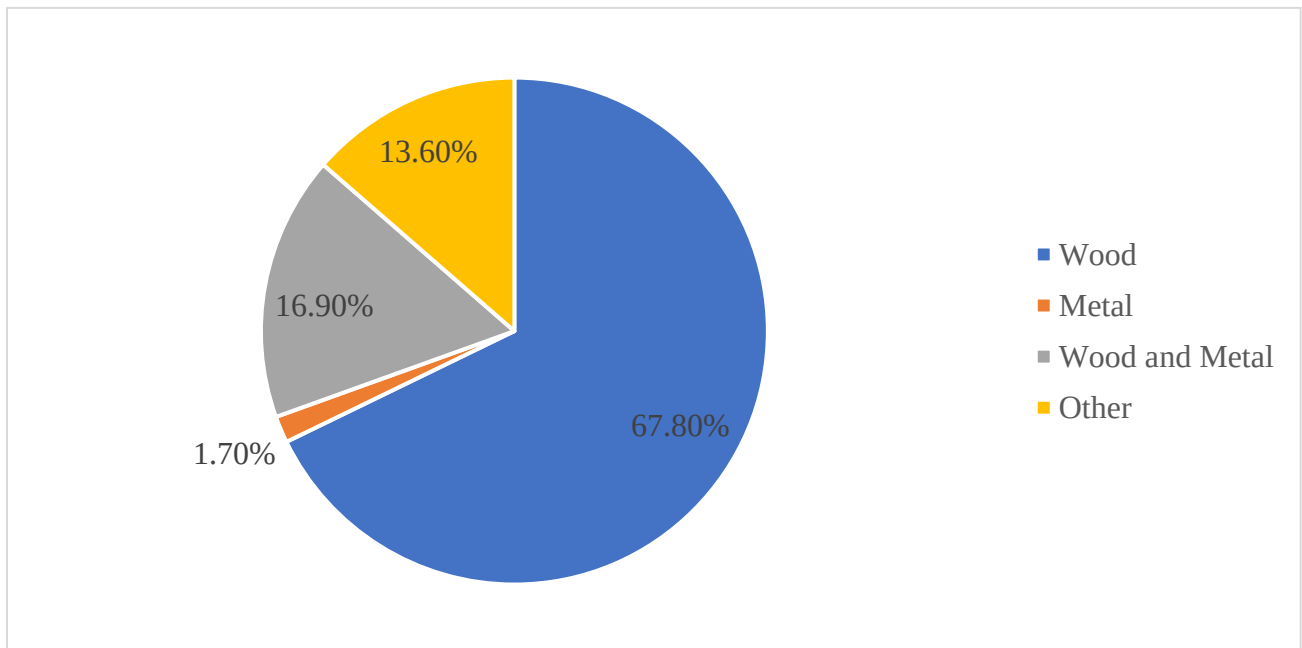


Figure 5: Pie Chart of Response count of respondents' study table material

6. Does your computer, unused books and stationary clutter your table?

- Yes
- No
- Maybe

Table 7: Frequency of respondents who feel that their table are cluttered

Options	Response Count	Response Percentage (%)
Yes	44	74.6
No	10	16.9
Maybe	5	8.5

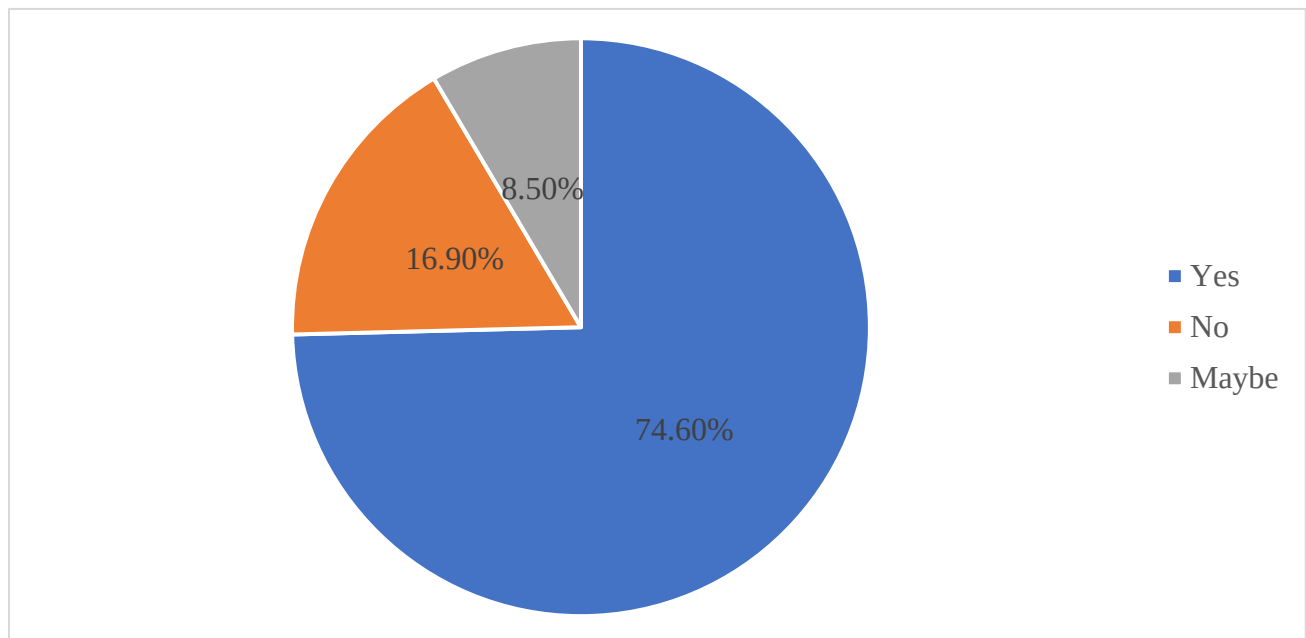


Figure 6: Pie chart of frequency of respondents who feel that their table are cluttered

7. Do you think a smart study table with pop up screen and shelves will be user friendly?

- Yes
- No
- Maybe

Table 8: Response count of user friendliness of pop-up screen and shelves

Options	Response Count	Response Percentage (%)
Yes	38	64.4
No	6	10.2
Maybe	15	25.4

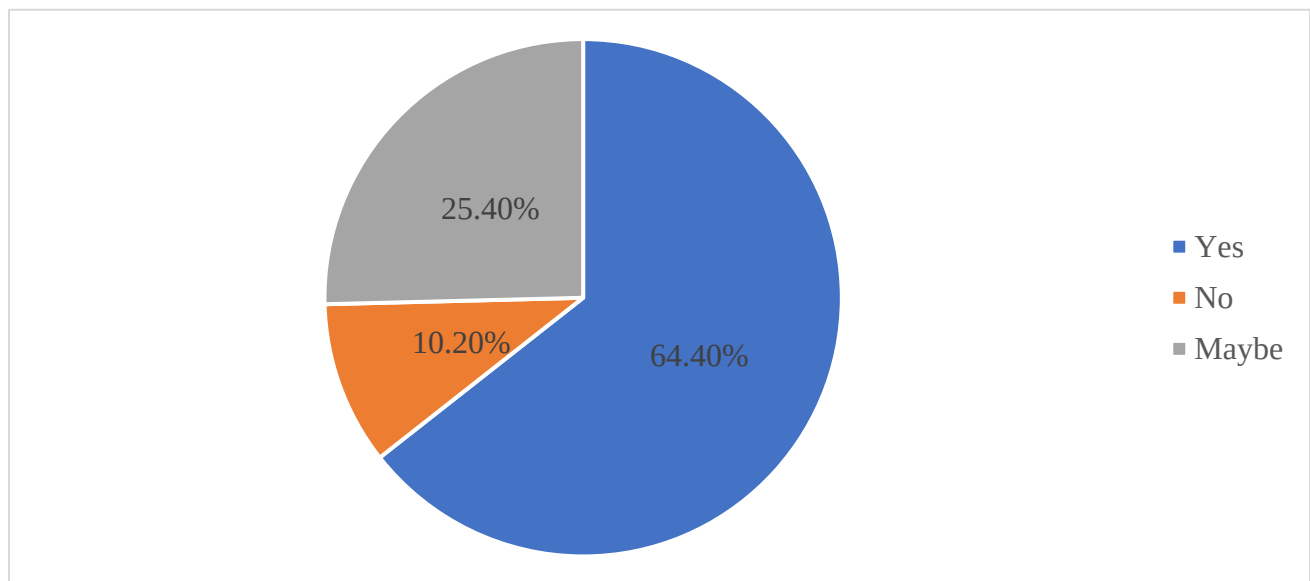


Figure 7: Pie chart of response count of user friendliness of pop-up screen and shelves

8. Would you prefer locking your table drawers with a computer?

- Yes
- No
- Maybe

Table 9: Response count of preference to lock drawers with a computer

Options	Response Count	Response Percentage (%)
Yes	35	59.3
No	15	25.4
Maybe	9	15.3

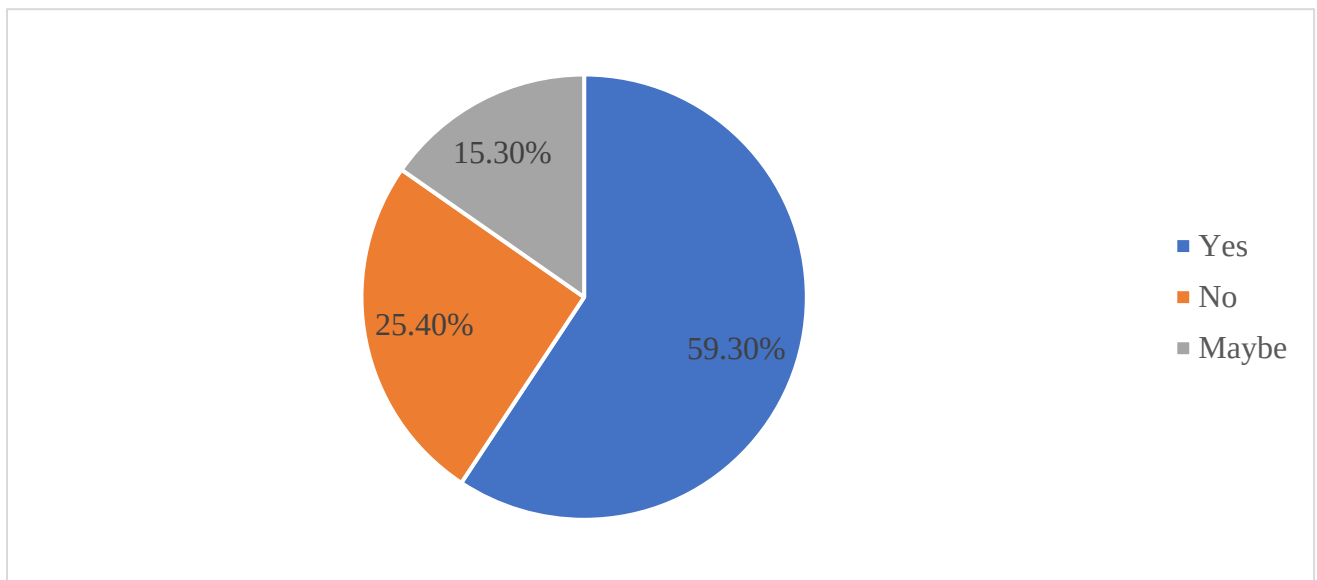


Figure 8: Pie chart of response count of preference to lock drawers with a computer

9. How large a computer screen would be sufficient for your daily activities?

- 9"
- 14"
- 17"

Table 10: Response count of preference of computer screen size

Options	Response Count	Response Percentage (%)
9"	7	11.9
14"	31	52.5
17"	21	35.6

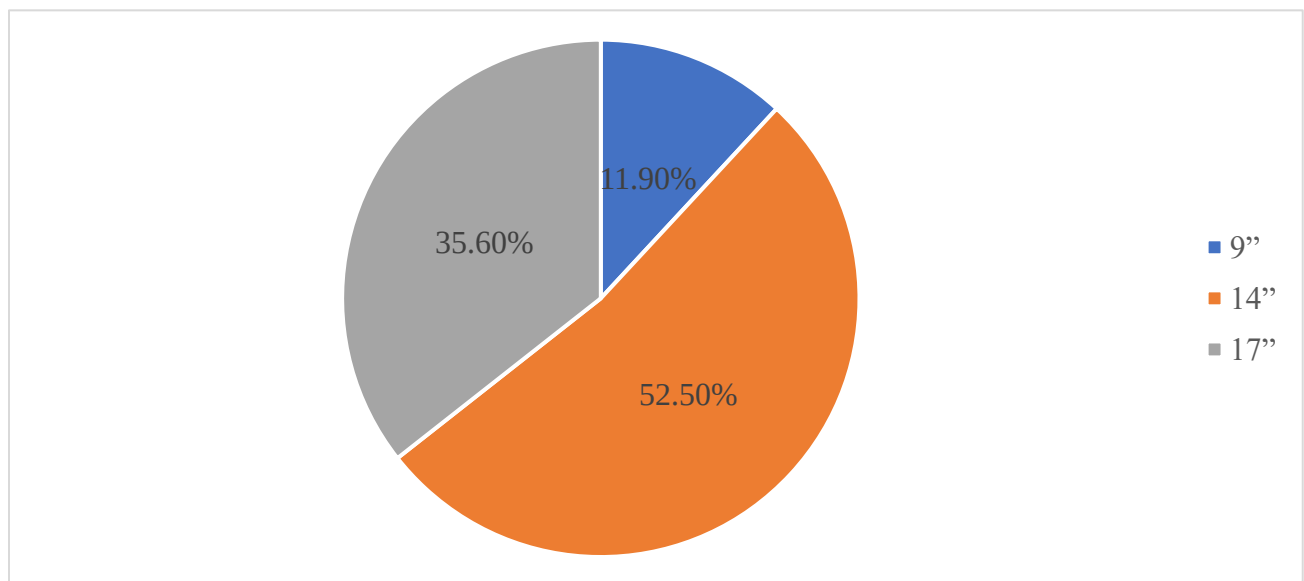


Figure 9: Pie chart of response count of preference of computer screen size

10. Would you prefer a portable table?

- Yes
- No

Table 11: Response count of preference of portability of the table

Options	Response Count	Response Percentage (%)
Yes	42	71.2
No	17	28.8

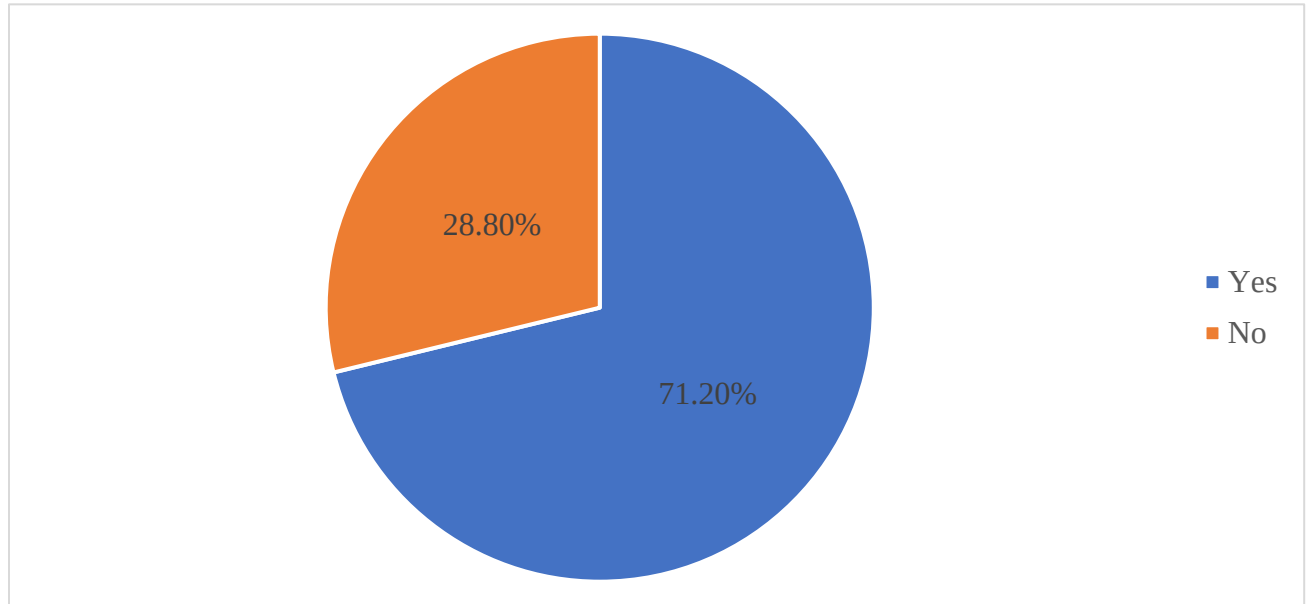


Figure 10: Pie chart of response count of preference of portability of the table

11. How long do you sit at your desk every day on average?

- 1-3 hours
- 3-5 hours
- 5 hours+

Table 12: Response count of daily average length of time respondents sit at their desks

Options	Response Count	Response Percentage (%)
1-3 hours	12	20.3
3-5 hours	18	30.5
5 hours+	29	49.2

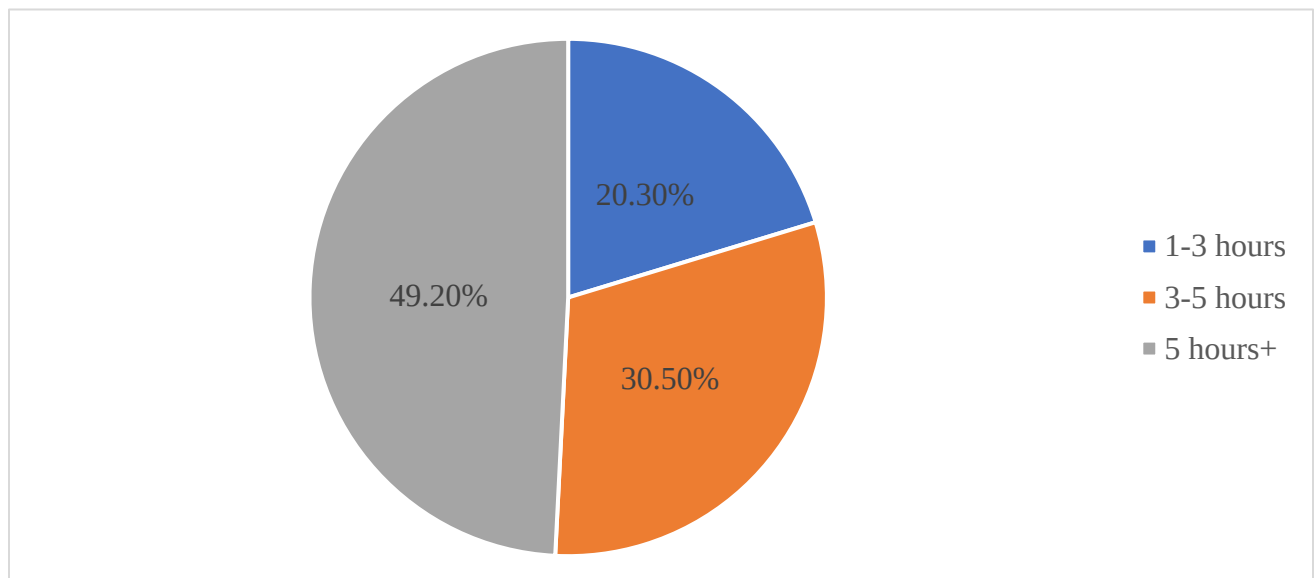


Figure 11: Pie chart of response count of daily average length of time respondents sit at their desks

12. Do you have back aches due to long study sessions?

- Yes
- No

Table 13: Response count of whether respondents get back aches or not

Options	Response Count	Response Percentage (%)
Yes	46	78
No	13	22

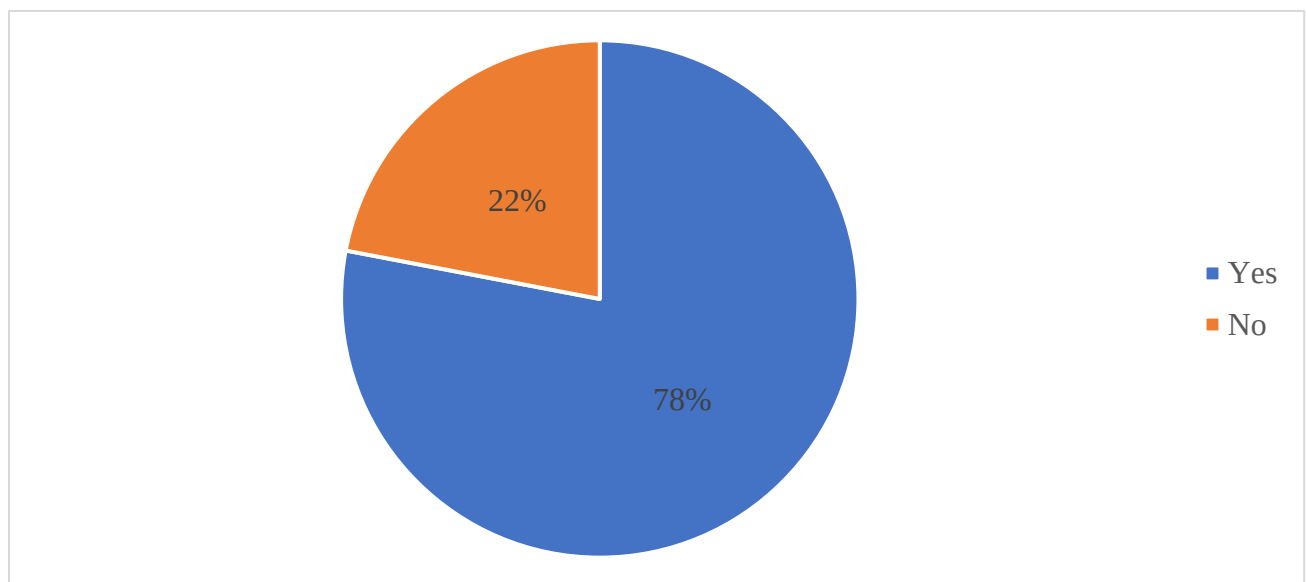


Figure 12: Pie chart of response count of whether respondents get back aches or not

13. How often do you have back aches due to long study sessions?

- Rarely
- Sometimes
- Frequently

Table 14: Response count of frequency of back aches of respondents

Options	Response Count	Response Percentage (%)
Rarely	17	28.8
Sometimes	29	49.2
Frequently	13	22

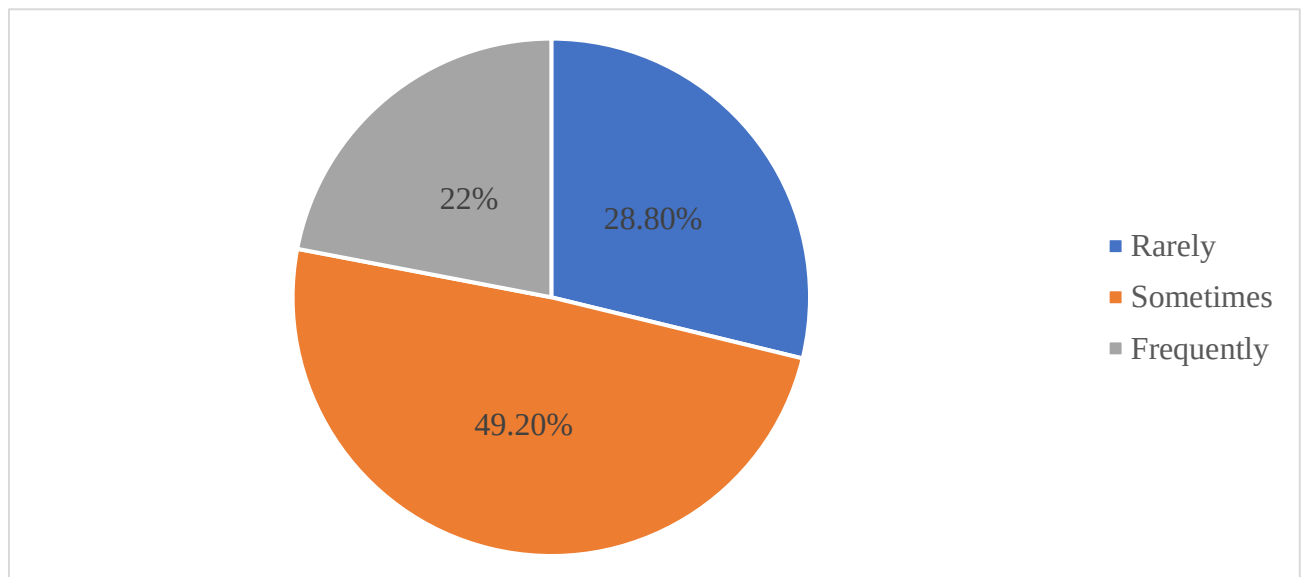


Figure 13: Pie chart of response count of frequency of back aches of respondents

14. Do you think standing up in between study sessions is beneficial in keeping good posture and decreasing back ache?

- Yes
- No
- I don't get back aches

15: Frequency of respondents who feel standing up between study sessions is beneficial

Options	Response Count	Response Percentage (%)
Yes	50	84.7
No	3	5.1
I don't get back aches	6	10.2

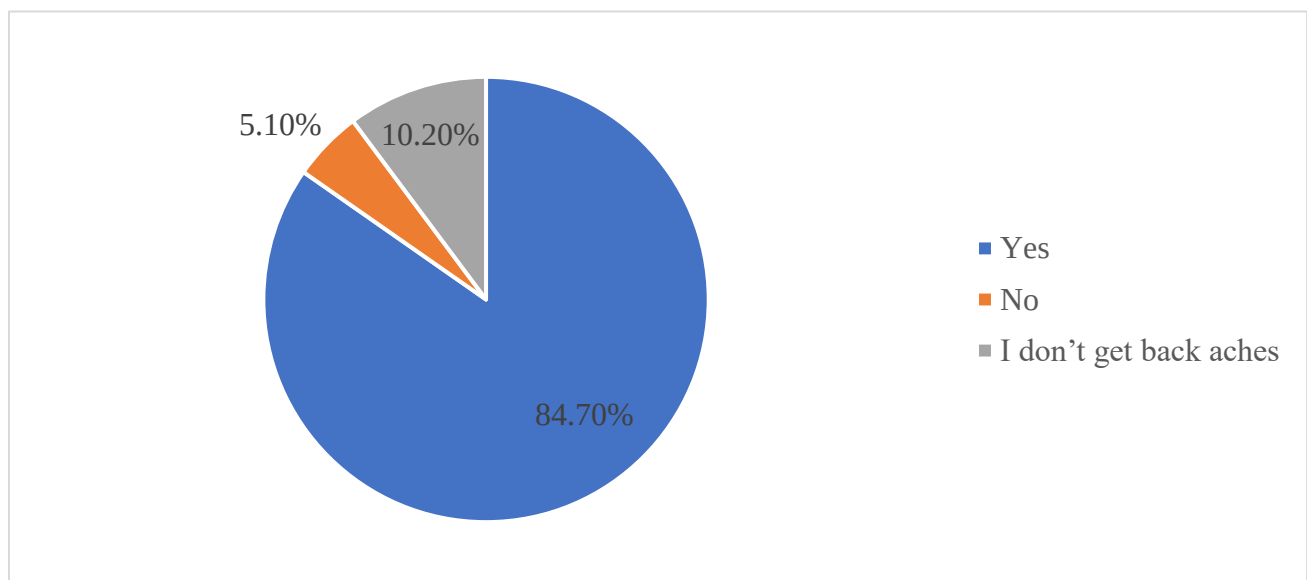


Figure 14: Pie chart of frequency of respondents who feel standing up between study sessions is beneficial

15. Would you prefer a table whose height can be adjusted so that you can work standing (to keep good posture and decrease back aches without sacrificing productivity)?

- Yes
- No
- I don't get back aches

Table 16: Frequency of respondents' preference of height adjustment of the table

Options	Response Count	Response Percentage (%)
Yes	50	84.7
No	5	8.4
I don't get back aches	4	6.9

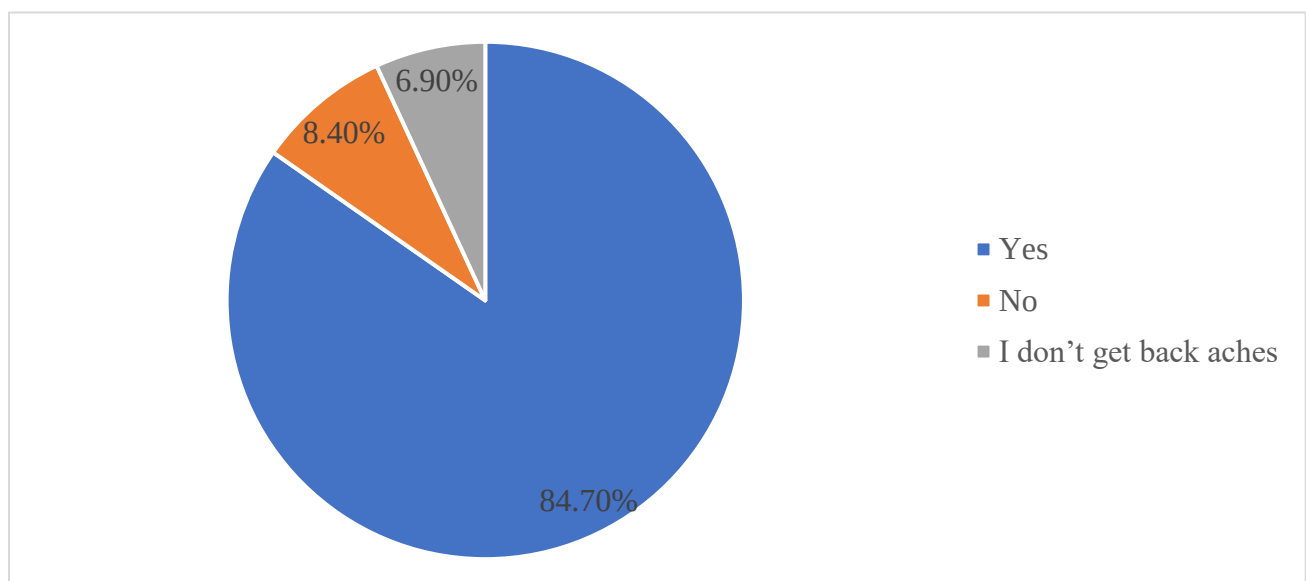


Figure 15: Pie chart of frequency of respondents who feel standing up between study sessions is beneficial

16. Would you prefer a mechanism that will adjust the computer screen height according to your eye level so as to reduce strain on your nape?

- Yes
- No
- Maybe

Table 17: Frequency of respondents' preference of adjusting computer screen height

Options	Response Count	Response Percentage (%)
Yes	45	76.3
No	4	6.8
Maybe	10	16.9

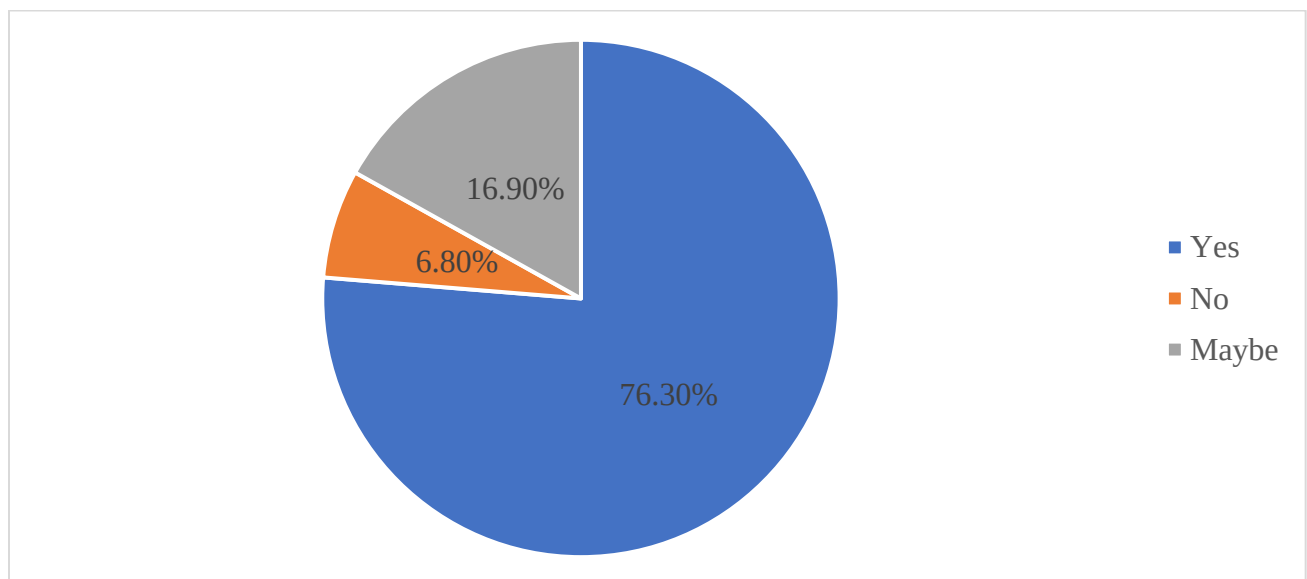


Figure 16: : Pie chart of frequency of respondents' preference of adjusting computer screen height

17. How comfortable will you be in switching from a conventional table to a smart study table?

- Very comfortable
- Somewhat comfortable
- Neutral
- Uncomfortable

Table 18: Frequency of respondents' comfort in switching to a smart study table

Options	Response Count	Response Percentage (%)
Very comfortable	21	35.6
Somewhat comfortable	23	39
Neutral	12	20.3
Uncomfortable	3	5.1

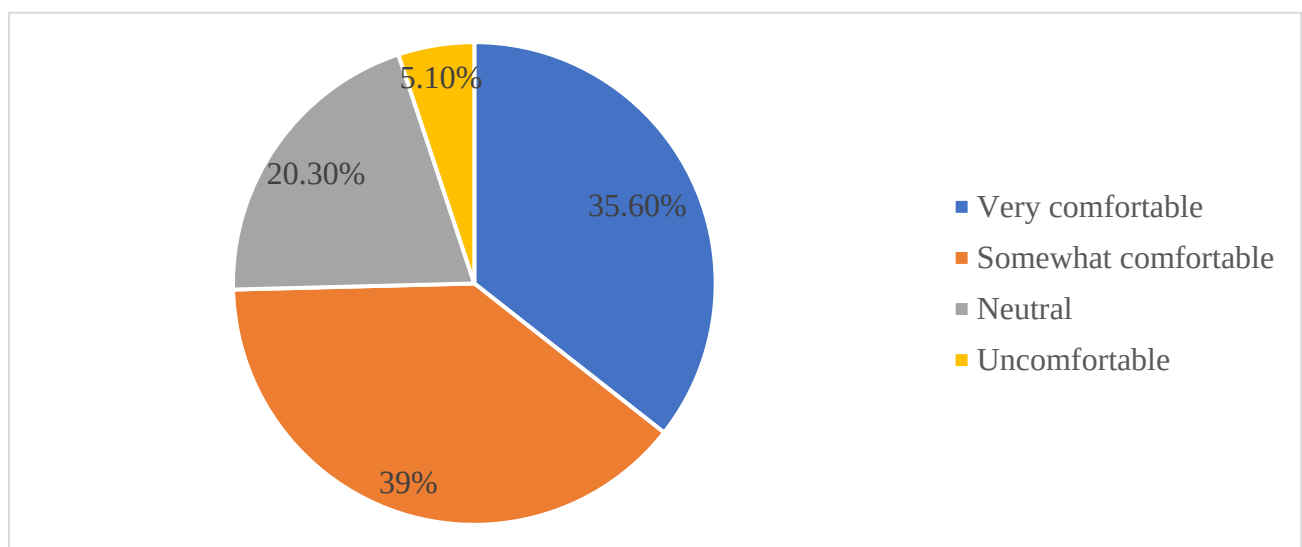


Figure 17: Pie chart of frequency of respondents' comfort in switching to a smart study table

18. What is your expected service life of this product?

- Less than 5 years
- 5-10 years
- More than 10 years

Table 19: : Frequency of respondents' expected service life of this product

Options	Response Count	Response Percentage (%)
Less than 5 years	12	20.3
5-10 years	30	50.8
More than 10 years	17	28.8

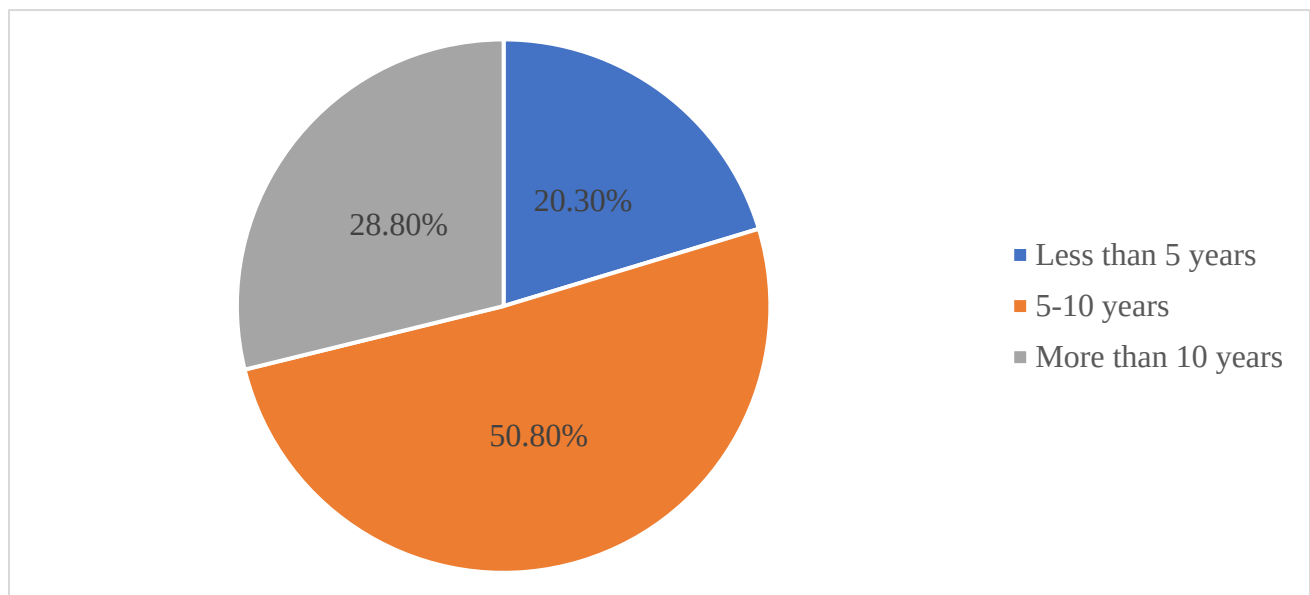


Figure 18: Pie chart of frequency of respondents' expected service life of this product

19. How much are you willing to pay for a smart furniture?

- 15,000-20,000
- 20,000-25,000
- 25,000-30,000
- 30,000+

Table 20: Frequency of respondents' expected price of a smart study table

Options	Response Count	Response Percentage (%)
15,000-20,000	38	64.4
20,000-25,000	13	22
25,000-30,000	3	5.1
30,000+	5	8.5

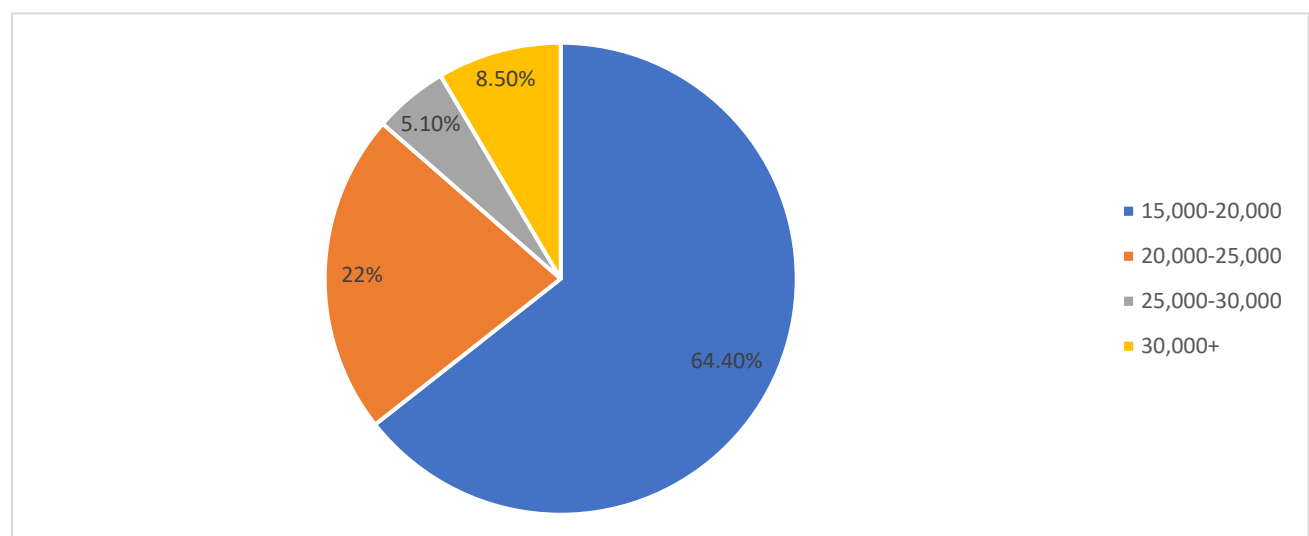


Figure 19: Pie chart of frequency of respondents' expected price of a smart study table

20. Which features are beneficial to you?

- Pop up shelves
- Height adjustment of monitor
- Height adjustment of table
- Separate pop-up computer section
- More table space and less clutter
- Computer controlled lock for drawers

Table 21: : Frequency of features that are beneficial to respondents

Options	Response Count	Response Percentage (%)
Pop up shelves	39	66.1
Height adjustment of monitor	36	61
Height adjustment of table	38	64.4
Separate pop-up computer section	29	49.2
More table space and less clutter	44	74.6
Computer controlled lock for drawers	29	49.2

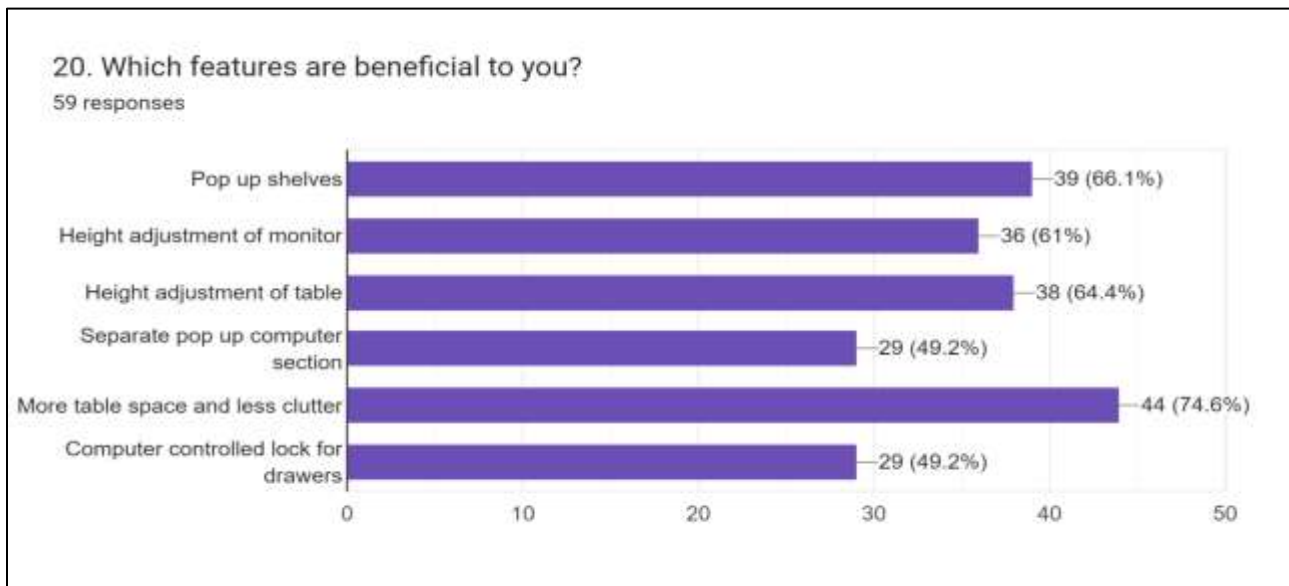


Figure 20: Bar chart of frequency of features that are beneficial to respondents

2.4 Customer's Requirement Evaluation

By evaluating the above questionnaires, we have given relative weight to different customer's requirements.

Table 22: Customer's Requirement and Their Relative Weight

Customers' Requirement	Relative Weight/ Importance (Scale of 10)
Flexibility	10
Comfort	10
Price	10
Security	9
Compatibility	9
Longevity	9
Portability	8
Material	8
Ease of maintenance	8
Branding	7

2.5 Conclusion

Surveys are carried out to collect customer requirements. From this customer requirements, we get a good idea of what the product design will look like. Questionnaires are meticulously developed and distributed through the mail, over the phone, or in face-to-face interviews in survey. Surveys are effective for gathering needs for redesigned or new products. Surveys let teams gather valuable insights they can use to generate user-centric product design ideas and prioritize the decisions that will boost conversions, customer satisfaction, and user engagement.

Chapter 3: Incorporating the Voice of Customer in Product Design with Quality Function Deployment (QFD)

3.1 Introduction

"Listen to the voice of the market" according to the Japanese Quality function deployment (QFD), which implies to "Understanding the design challenge." To create a high-quality product, one must first understand the design challenges. The translation of user requirements and demands into a technical description of what has to be created, emphasized or neglected is known as quality function deployment (QFD). Instead of creating a solution that highlights the builder's existing knowledge, QFD's purpose is to produce a product that performs precisely what the client wants. Developing "technical requirements" is an even more challenging task than identifying the correct problem to tackle. Engineering specifications are written using a variety of methods. Multi-skilled teams work together in a QFD process to get a shared understanding of the customers' demands and define the necessary technical requirements for each step. The main benefit of the QFD technique is that it is arranged to produce the major pieces of knowledge required to comprehend the problem:

1. Paying attention to the consumers' voices
2. Improving teamwork
3. Improving production efficiency
4. Processes that are driven by customers rather than by technology
5. Shortening the time and cost of development
6. Specifying the requirements
7. Examining how the specifications match the needs of the consumers
8. Assessing how successfully the competition achieves its objectives
9. Identifying goals and objectives to strive toward

After gathering customer demands via a survey, the following stage in the QFD approach is to assess the relevance of each of the customers' wants (out of 10 scale). This would be done by calculating a weighting factor.

3.2 Quality Function Deployment's House of Quality

Since we have already collected customer needs by conducting a survey, the next step in the QFD technique is to evaluate the importance of each of the customers' requirements (out of 10 scales). This is accomplished by generating a weighting factor. After rating each parameter, a gross score is calculated which is in this case equal to the important factor.

Table 23: Relationship Explanation

Customer Requirements	Requirement	Relationship	Explanation
Flexibility	Pop Up Mechanism	Weak	Pop up mechanism will increase space utilization.
	Wheels	Weak	Wheels adds portability
	Computer controlled locking module	Moderate	It will ensure easy to use security system.
	Auto Adjustment Module	Strong	Self-adjustment will add more comfort to the user
Comfort	Pop Up Mechanism	Weak	It will increase space utilization.
	Active Sensors	Moderate	It will follow the human motion and respond accordingly.
	Computer controlled locking module	Strong	It will relieve the users from locking up the doors physically and keeping multiple keys.
	Auto Adjustment Module	Strong	It will add flexibility to the user.
Price	Malaysian MDF wood	Strong	Quality of raw plywood determines the major costs.
	Computer system	Strong	Specifications of computer used contribute much to the total cost.
	Computer	Strong	Locking modules are available but

	controlled locking module		quite costly.
	Accessories	Weak	Accessories are cheaper.
	Auto Adjustment Module	Moderate	Auto adjustment module has also added high manufacturing cost.
Security	Computer controlled locking module	Strong	It will ensure easy to use security system.
	Malaysian MDF wood.	Weak	Customers' most preferred material for table is wood. So, we do not have much things to do with material
Compatibility	Computer controlled locking module	Strong	It may take time for the users' to cope up with this mechanism.
	Auto Adjustment Module	Strong	Flexibility mainly depends on it.
	Pop Up Mechanism	Moderate	Some of the users might not find this feature helpful.
	Accessories	Weak	They will just add more convenience to use.
Durability	Pop Up Mechanism	Strong	This is an electrically controlled linearly actuating mechanism.
	Active Sensors	Strong	These sensors are very sensitive and keep much impact on Longevity.
	Computer controlled locking module	Strong	These modules are not that much available in our country.
	Auto Adjustment Module	Strong	This is one of the core parts of the whole system.

	Malaysian MDF wood.	Strong	It is the main component.
	Computer system	Strong	Whole mechanism will be controlled by this central computer.
	Accessories	Weak	They are pretty much available and long lasting.
Portability	Wheels	Strong	Wheels are must for portability
	Malaysian MDF wood.	Strong	Low price wood is weak and very heavy.
	Accessories	Weak	Accessories are not that much heavy.
Material	Malaysian MDF wood.	Strong	Plywood is the main building component.
Ease of Maintenance	Malaysian MDF wood.	Weak	Plywood has enough strength to confront the circumstances that it will go through.
	Accessories	Moderate	They have much longevity and are easy to be maintained.
	Computer system	Strong	Preventive maintenance is required.
	Auto Adjustment Module	Strong	Regular and Preventive maintenance is required.
	Computer controlled locking module	Strong	It also requires systematic maintenance.
	Active Sensors	Strong	Its maintenance is very sensitive.
	Pop Up Mechanism	Moderate	It is not comprised with any component that is hard to be maintained.
	Wheels	Weak	They do not require any routined maintenance.

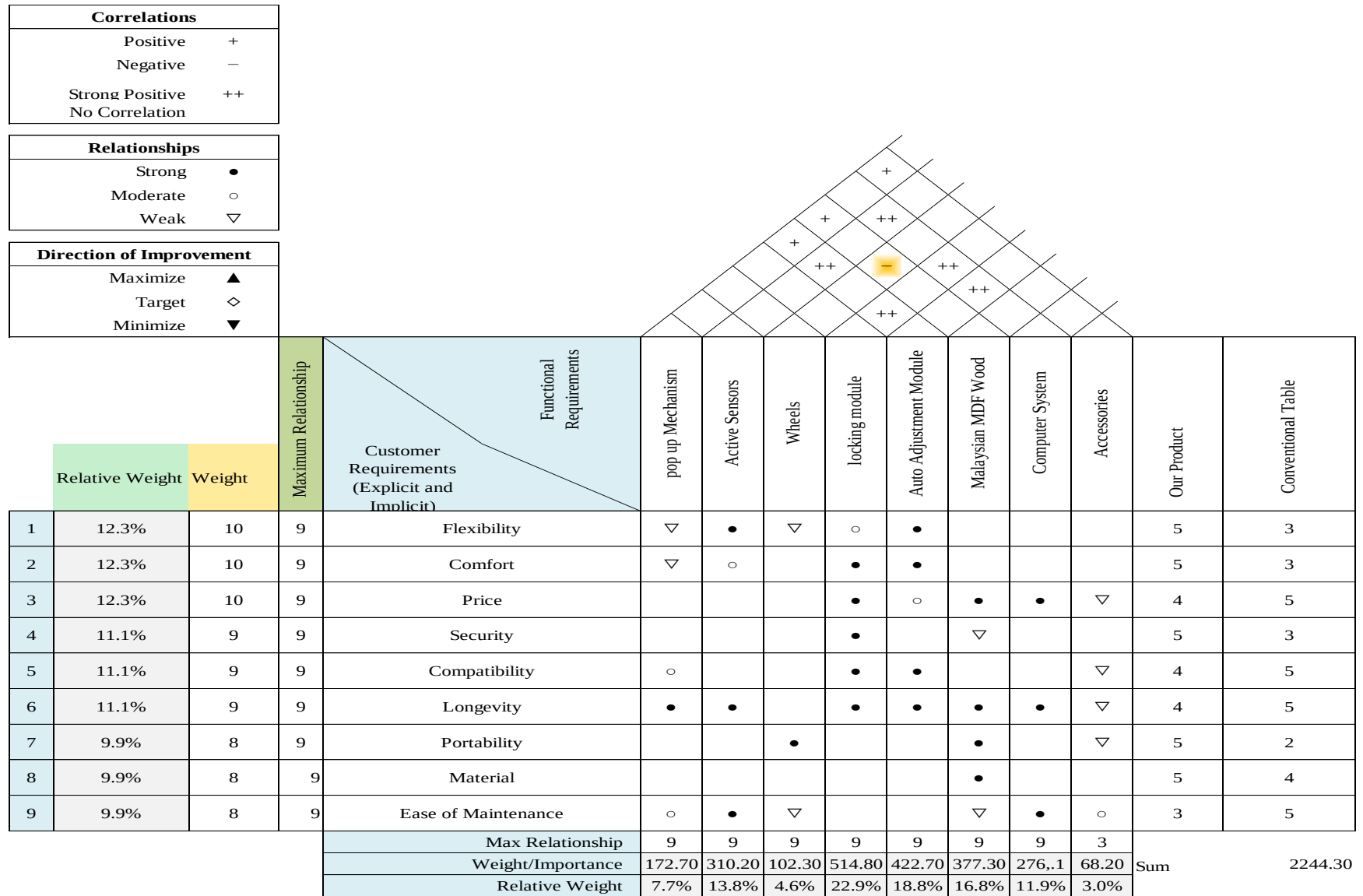


Figure 21: QFD

3.3 Conclusion

The QFD method ensures that the issue is well comprehended. It helps companies reach a consensus on measuring techniques and performance criteria that will satisfy customer needs and converts these needs to technical substitutes. Its goal is to help a firm become more strategically competitive. It also prioritizes the measures that must be taken to meet the customers' spoken and unspoken needs. Due to a shortage of time, we had to build QFD quickly. However, QFD is a time-consuming and systematic procedure. It may seem to slow down the design process, but it does not, since the effort spent on creating knowledge now is offset by time saved later on.

Chapter 4:Functional Decomposition

4.1 Introduction

Functional decomposition is a technique for dividing a larger task into smaller tasks. In the process of product design and development, a proper sequence of methods, tools and techniques are to be applied. Large or complex functionalities are more easily understood when broken down into pieces using functional decomposition. During decomposition, the team focuses on the inputs and outputs of each task and its interaction between various components. The decomposition process is usually applied recursively until the design team can readily envision ways of achieving each of the sub-tasks. Functional decomposition is also called functional analysis, top-down design, or logical decomposition. Black box analysis is very handy tool to address this issue. The product is simply considered as a black box the mechanism of which is unknown. Energy conversion steps are very important in this regard. As soon as the functionalities expected from the product are set, a detailed analysis of the product's infrastructure is to be made. Component hierarchy shows a simple yet effective layout of components to achieve the required functionality.

Four basic steps are usually followed for a successful decomposition. They are given below.

Step 1: Find the most general function

In this step, the most general task of the design that needs to be accomplished is identified. This is usually a one-sentence answer. A black box diagram is usually the output of this step where materials, energy, information inputs, and outputs are specified.

Step 2: Find the closest sub-functions

The functions that must take place immediately close to the most general function are specified here.

Step 3: Find the next level(s) of sub-functions

For each of the functions that are specified in step 2, the closest sub-functions are found and they are drawn on the functional decomposition diagram.

Step 4: Refine Sub-functions

It is checked that there are no additional functions that are not included. If there are any they are put in the right place.

4.2 Black Box Model of Functional Decomposition

A black box systems model is used to identify what goes into and out of the product in order to make it work as a system. A black box model is more useful way to perform the functional decomposition. In this model, the product is modeled abstractly as a black box with inputs and outputs. The flow of inputs (material, energy, and information) to outputs is sufficient to describe a technical system or product. Black box diagram of Smart Pop-up Table is shown below:

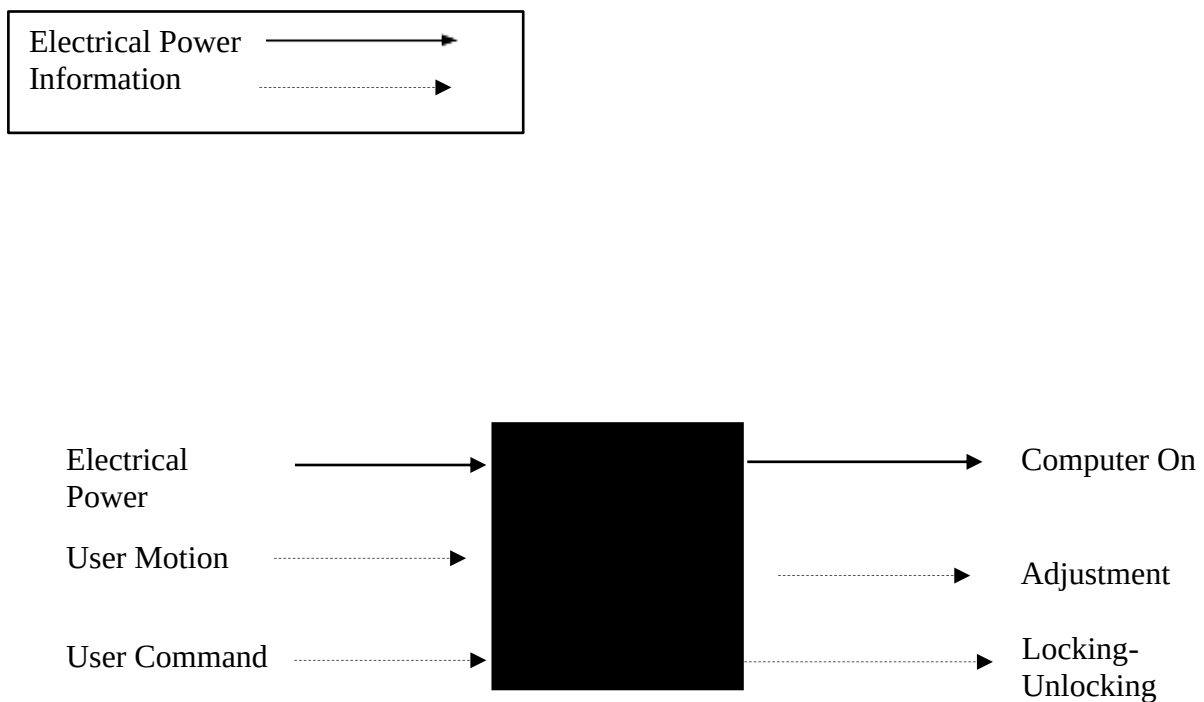


Figure 22: Black Box Model of Smart Pop Table

4.3 Component Hierarchy

Component hierarchy is the classification of a product into its essential components. Component hierarchy is a very effective method of listing the components required to design a product. An elementary approach to functional decomposition is to decompose the prime function hierarchically into sub functions; when all sub functions are satisfied, the prime function is satisfied. This can be repeated iteratively down several levels developing a function tree. Function trees are fast and easy to construct, but this ease of construction comes at the expense of understanding interactions between sub- assemblies.

The component hierarchy of our Smart Pop-up Table is shown on the following page:

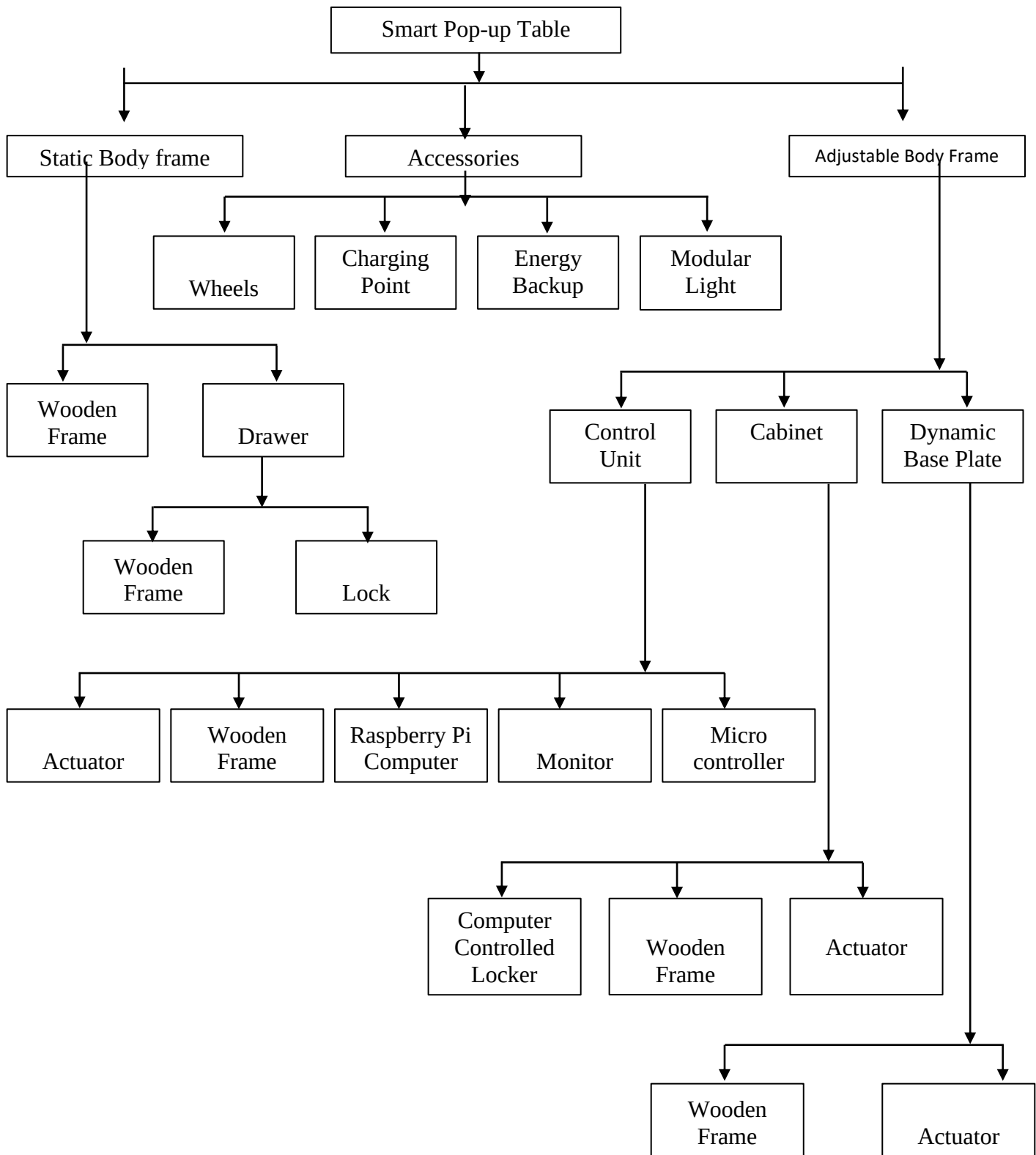


Figure 23: Component hierarchy of Smart Pop-Up Table

4.4 Cluster Function Structure for Smart Pop-Up Table

The cluster function shows the flow of energy, main working principle. The three cluster functions have been shown here

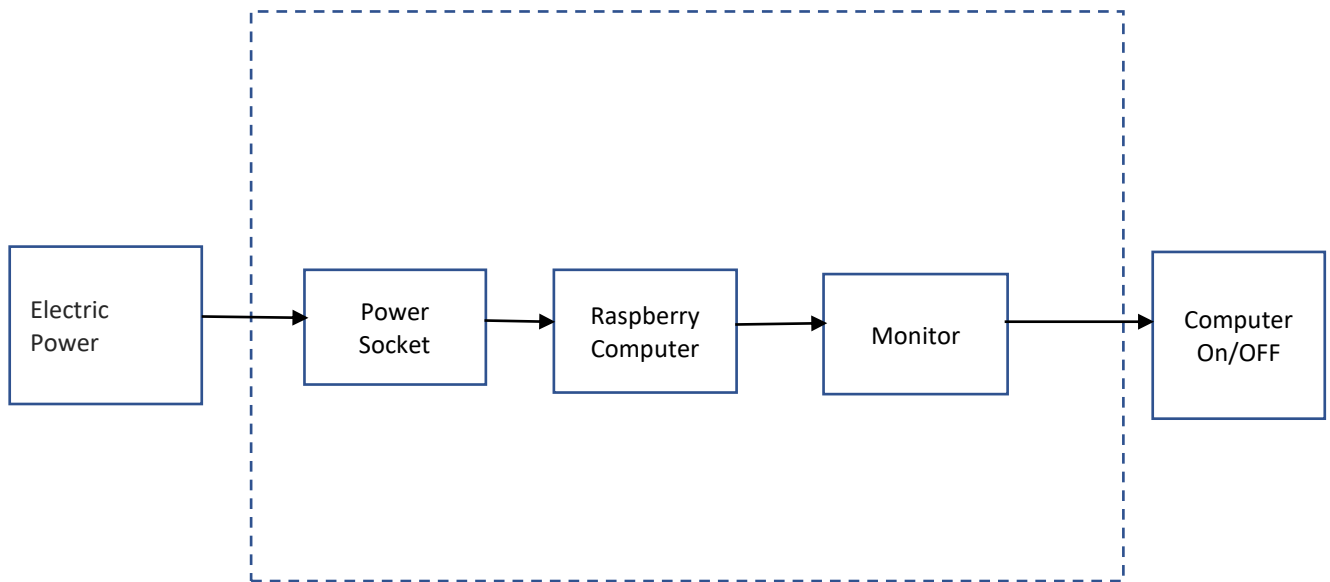


Figure 24: Cluster Function of electric power

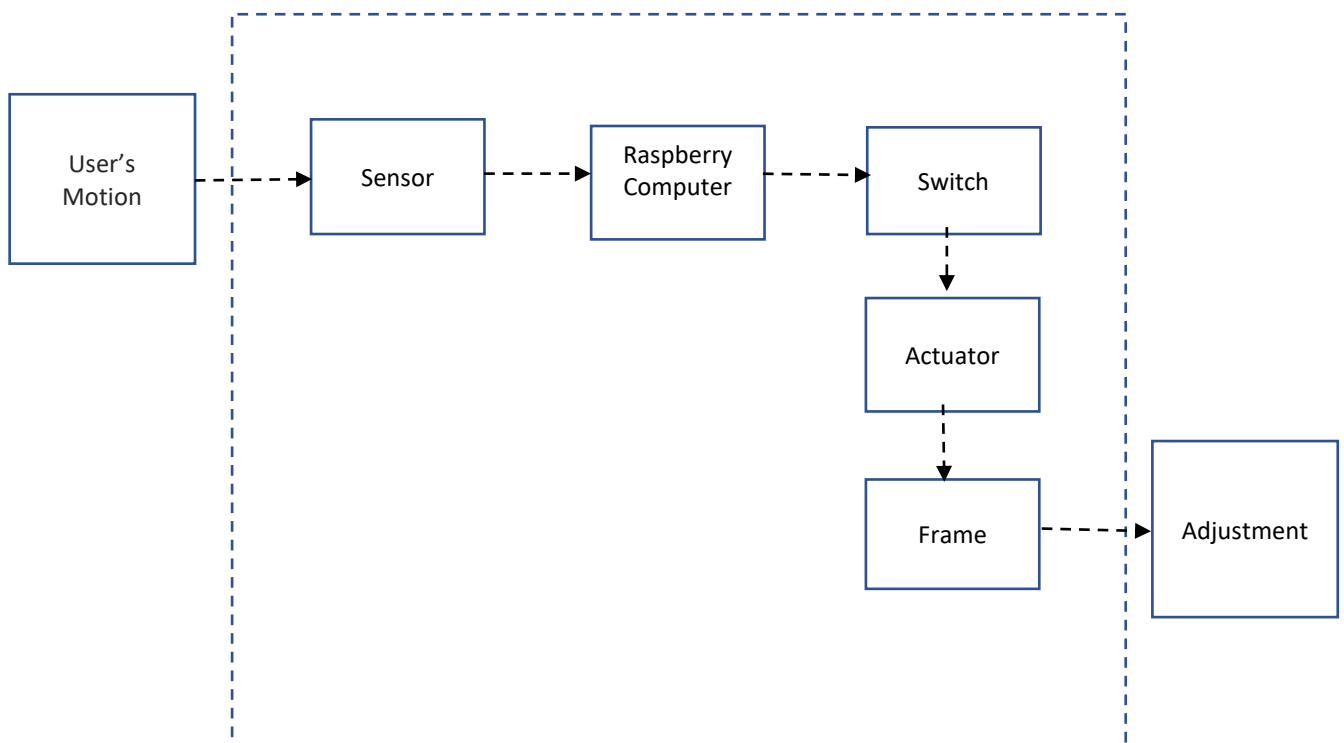


Figure 25: Cluster Function of user's motion

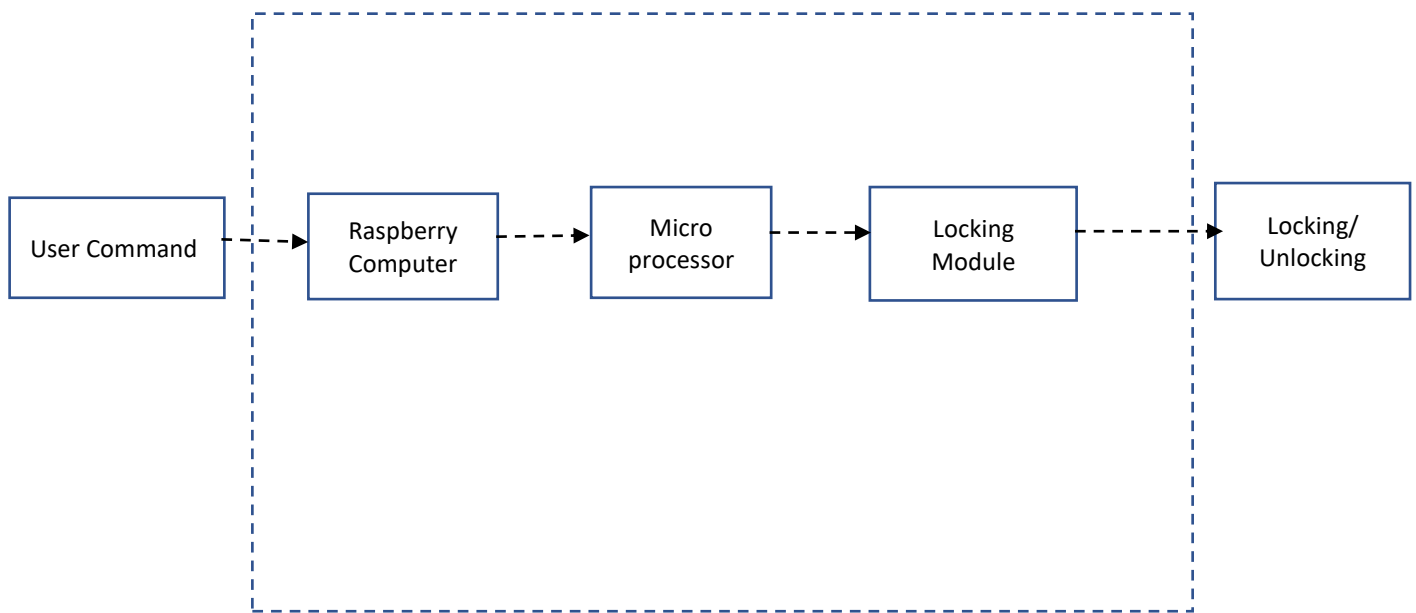
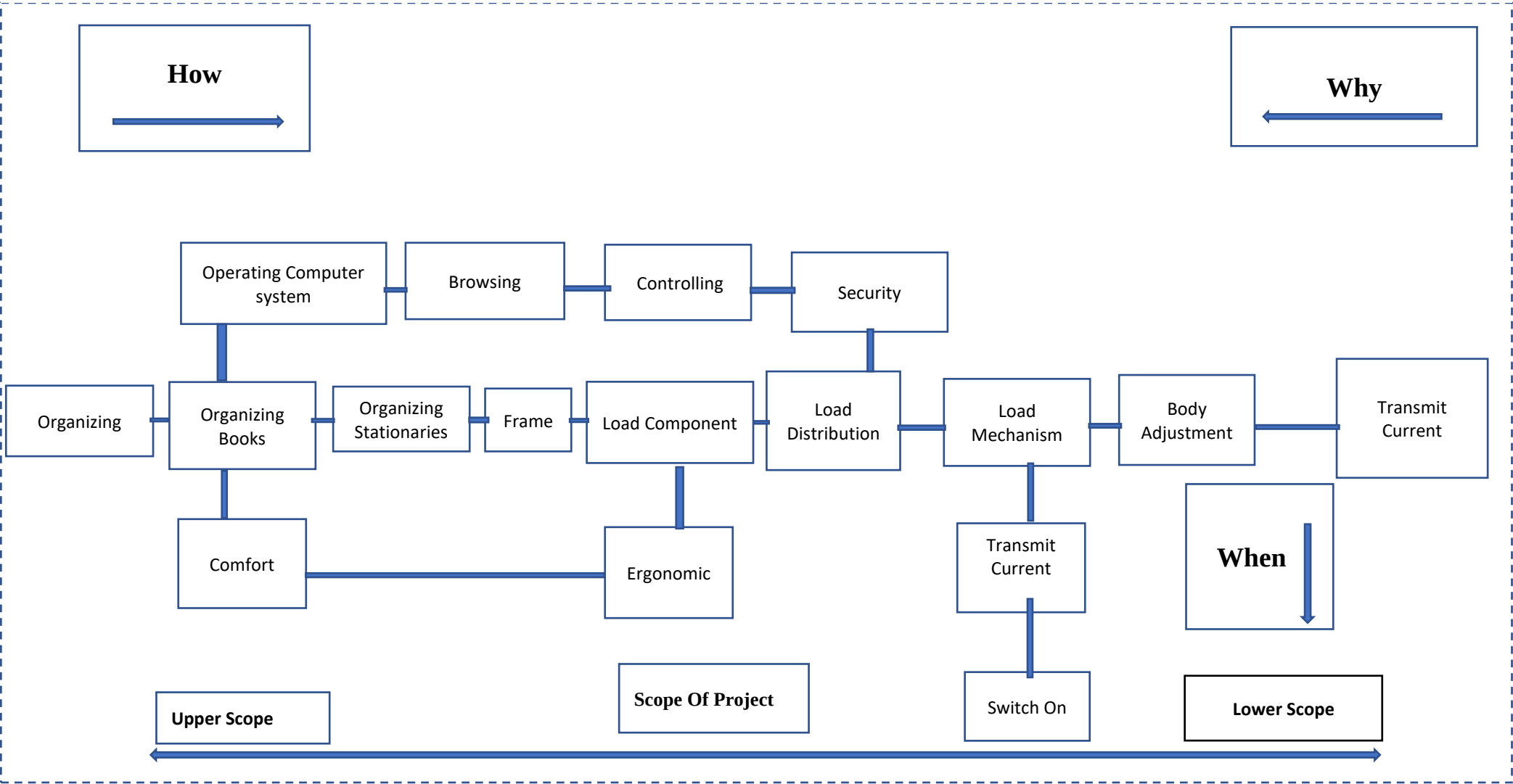


Figure 26: : Cluster Function of user command

4.5 FAST Diagram

Figure 27: Fast Diagram



4.6 Conclusion

Functional decomposition is of the utmost importance in architecting the system. A functional decomposition becomes increasingly important as complexity increases. As the name says “BLACK-BOX”, what happens inside the black-box is not visible. The component hierarchy for different mechanism shows hierarchy of the levels of increasingly detailed processes within the system and the cluster function structure show the flow of material, energy and information of the whole product with the help of a detail drawing. The black box model, Component hierarchy, and Cluster function structure, these three methods are very useful ways for generating a successful functional decomposition.

Chapter 5: Design Analysis

5.1 Introduction

The systematic process of developing a design, which includes all information discovery, planning, and communications, is known as design analysis. A product's development is divided into various phases, each of which entails a set of actions. Most of the tasks need various sorts of analysis that concentrate on various areas of the company, such as market analysis to assess prospective consumers' buying behavior, financial analysis to determine potential risks and investment possibilities, and so on. Some design assessments, on the other hand, are required for the new product to become a reality.

One of the most important phases of product development is design analysis. Physical prototyping and testing may be reduced, if not eliminated entirely. A prototype design of a “Smart Pop-Up Table” has been produced after all of the study of client demands, material, manufacturing process, and joining technique selection. In this chapter, all of the components are presented, which were developed using SolidWorks software, and then the stress analysis of the “Smart Pop-Up Table” is illustrated using Ansys software.

5.2 Final Design & Parts of the “Smart Pop-Up Table”

5.2.1 Parts of “Combined Tiller and Sprayer”



Figure 28: Smart Table Frame



Figure 29: Monitor Frame



Figure 30: Drawer

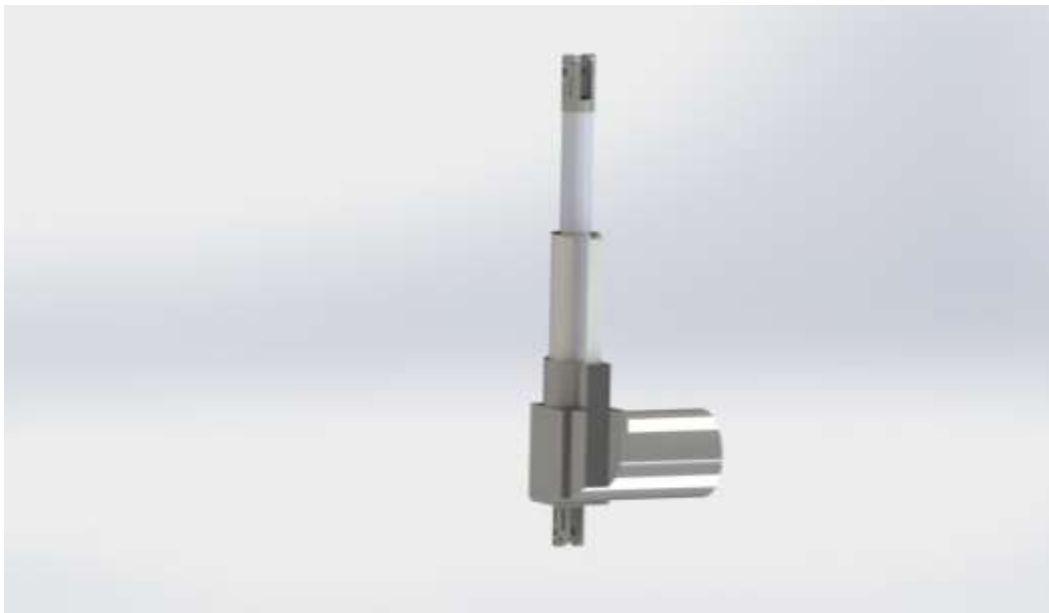


Figure 31: Linear Actuator

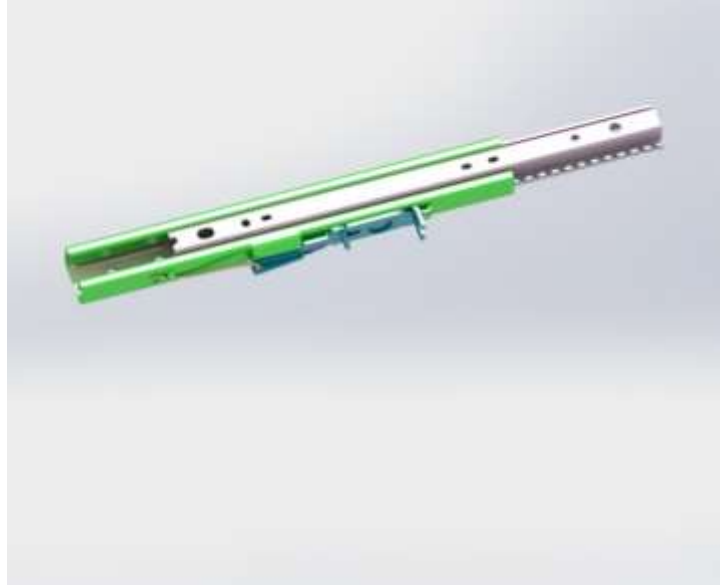


Figure 34: Drawer Channel

5.2.2 Final Design



Figure 35: Final Design of Smart Table

5.3 Static Structural Analysis (ANSYS)

5.3.1 Static Structural Analysis of Static Frame (Lower Frame)

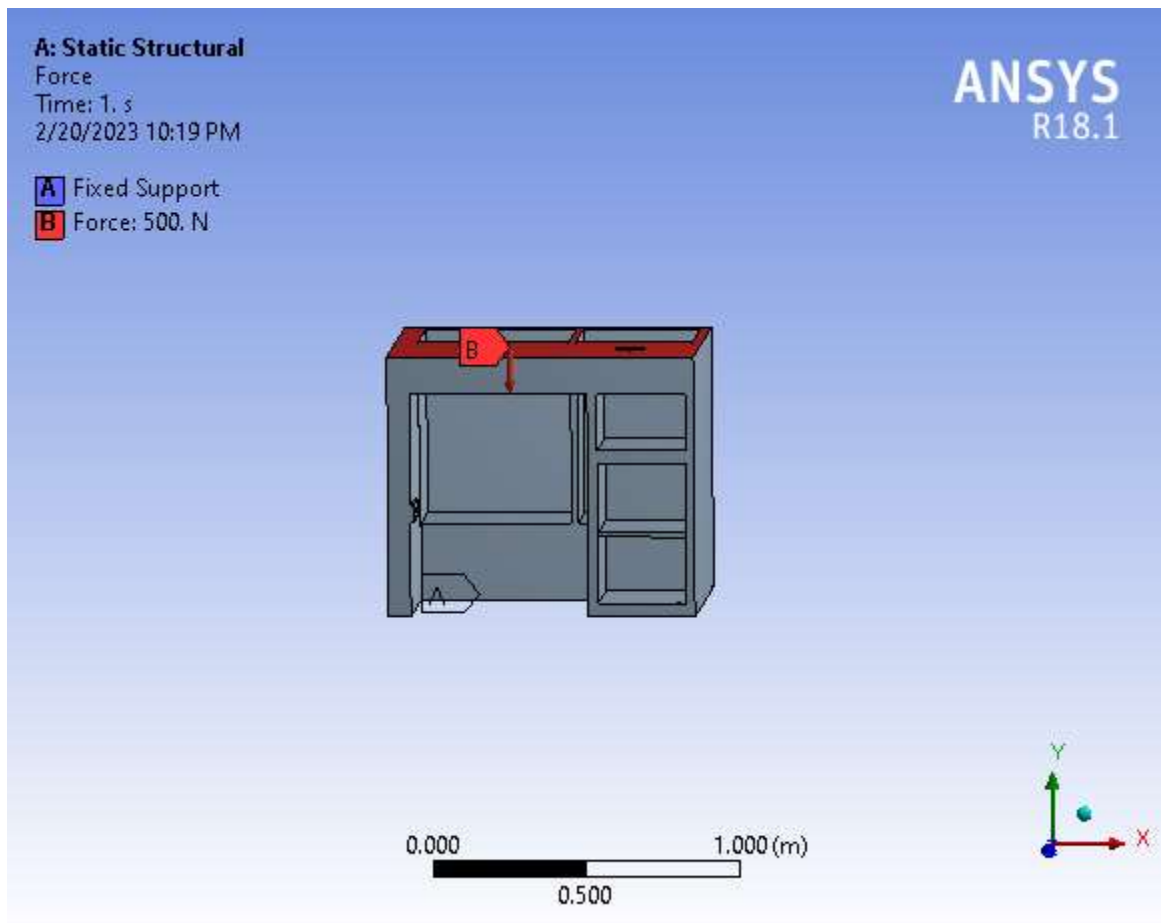


Figure 36: Static Structural Analysis of Static Frame (Lower Frame)

5.3.1.1 Total Deformation of Static Frame (Lower Frame)

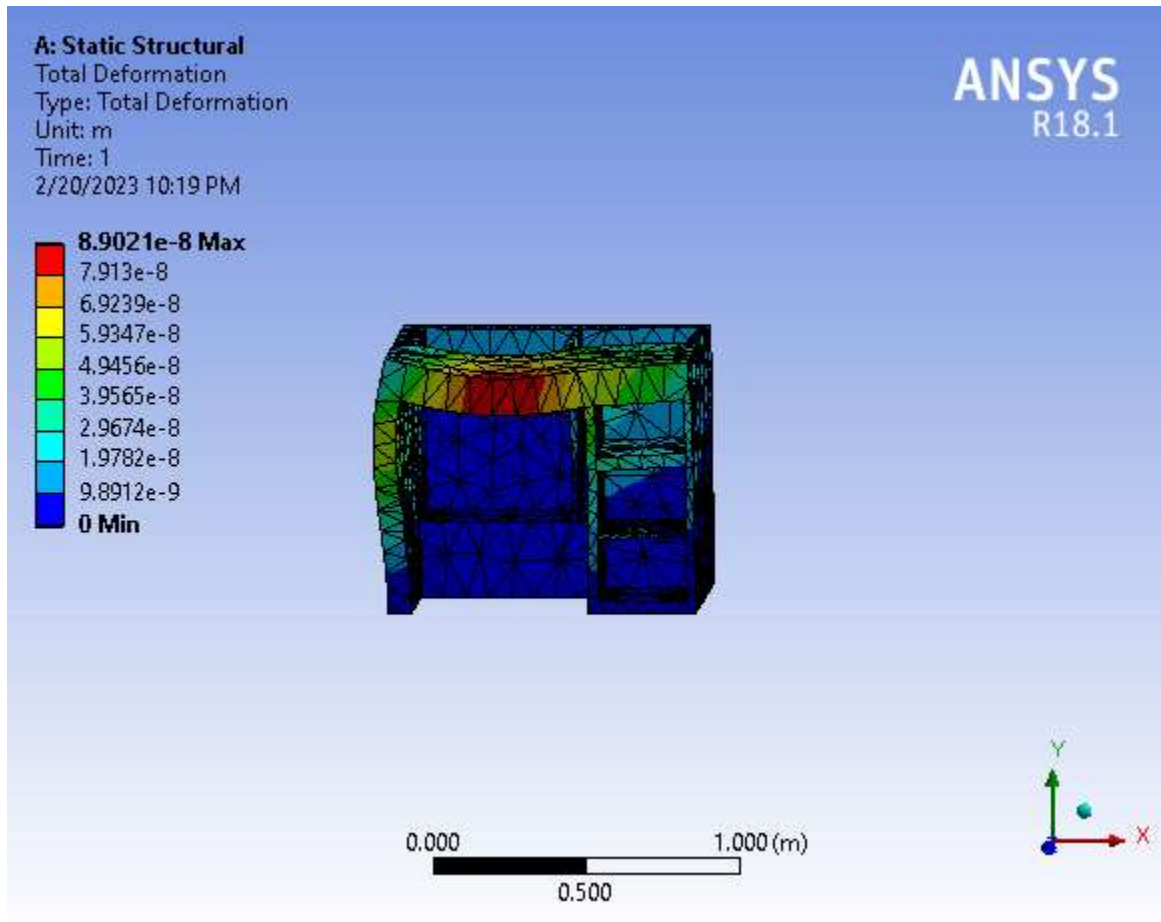


Figure 37: Total deformation of Static Frame (Lower Frame)

Table 5.1: Total deformation of Static Frame (Lower Frame)

	Total Deformation (m)
Minimum	0
Maximum	8.9021×10^{-8}

5.3.1.2 Equivalent (Von-Mises) Stress

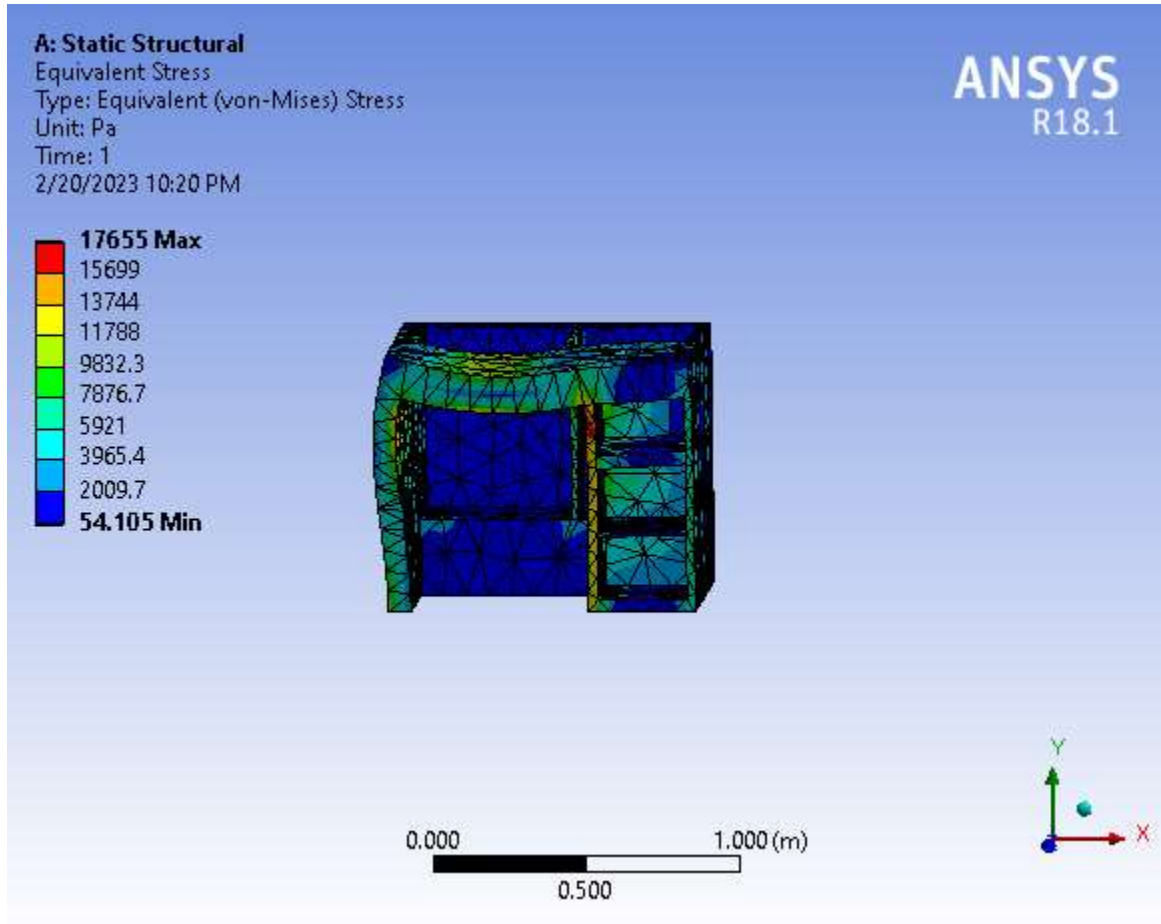


Figure 38: Equivalent (Von-Mises) Stress of Static Frame

Table 24: Equivalent (Von-Mises) Stress of Static Frame

	Equivalent (Von-Mises) Stress (Pa)
Minimum	54.105
Maximum	17655

5.3.1.3 Equivalent Elastic Strain of Static Frame

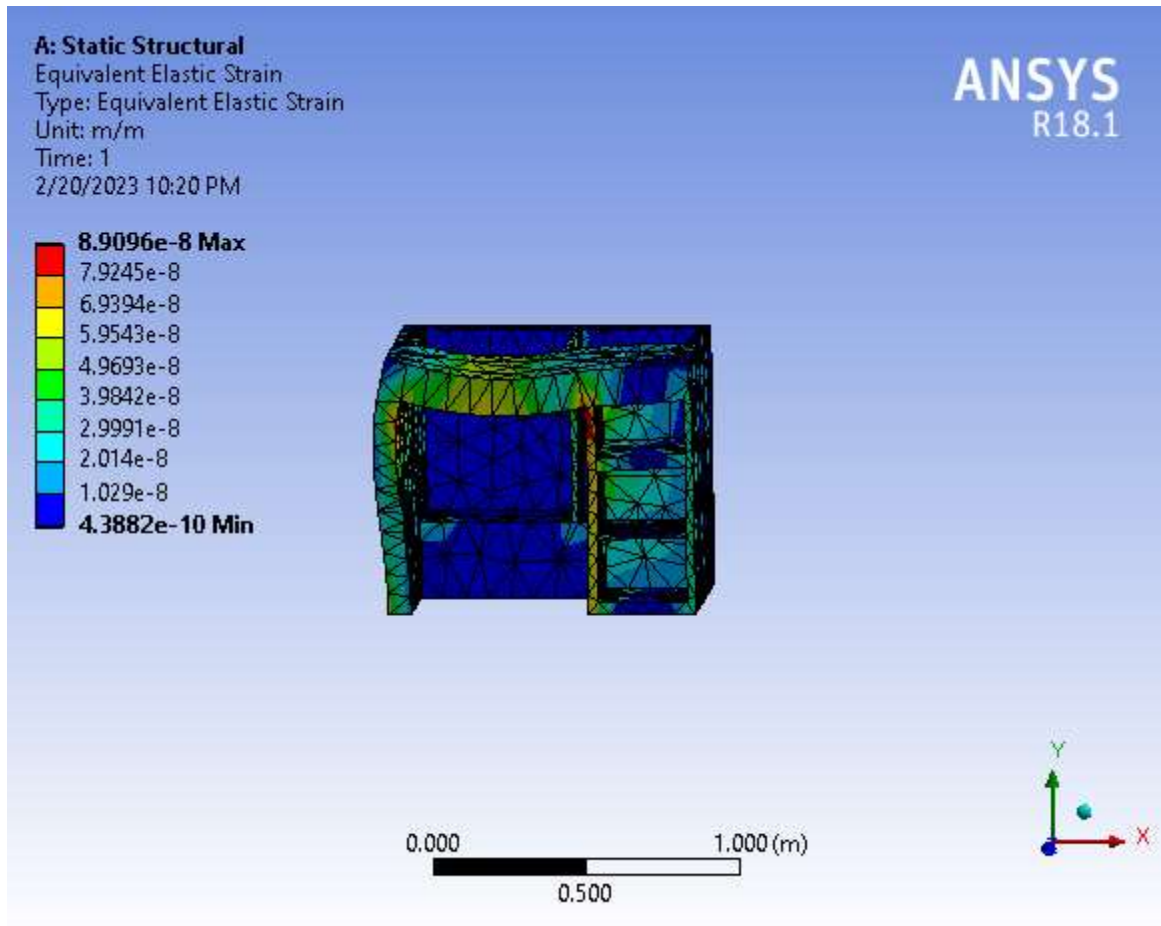


Figure 39: Equivalent Elastic Strain of Static Frame

Table 25: Equivalent Elastic Strain of Static Frame

	Equivalent Elastic Strain of Static Frame (m/m)
Minimum	4.3882×10^{-10}
Maximum	8.9096×10^{-8}

5.3.2 Static Structural Analysis of Linear Actuator

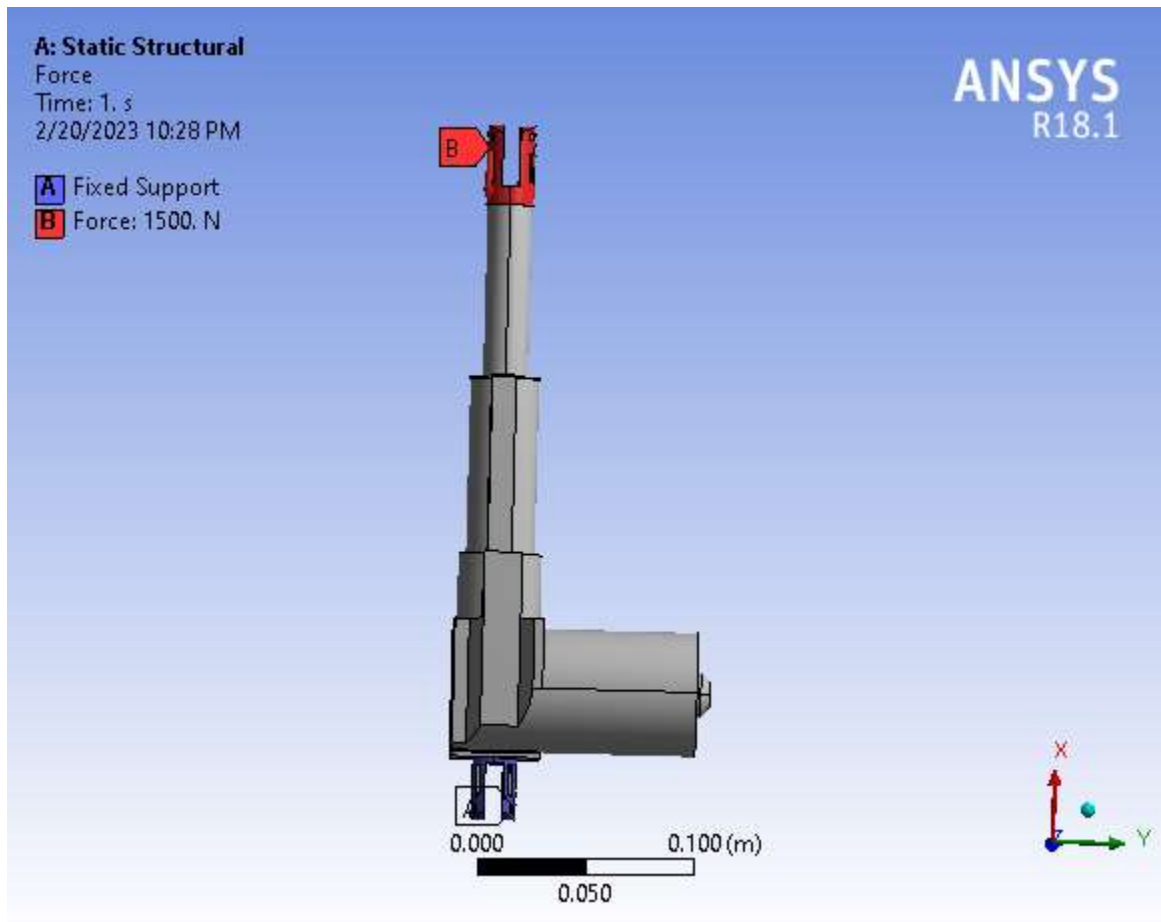


Figure 40: Static Structural Analysis of Linear Actuator

5.3.2.1 Total Deformation of Linear Actuator

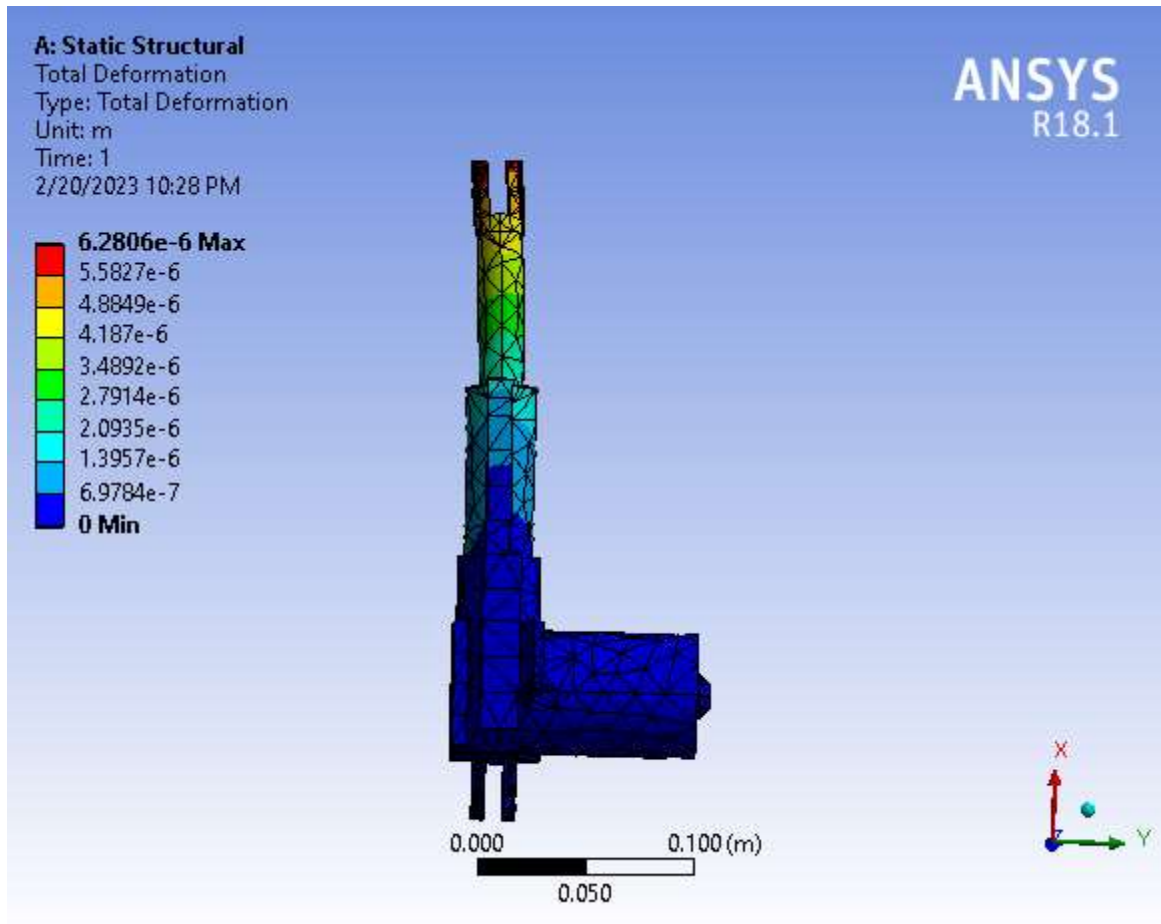


Figure 41: Total deformation of Linear Actuator

Table 26: Total deformation of Linear Actuator

	Total Deformation (m)
Minimum	0
Maximum	6.2806×10^{-6}

5.3.2.2 Equivalent (Von-Mises) Stress

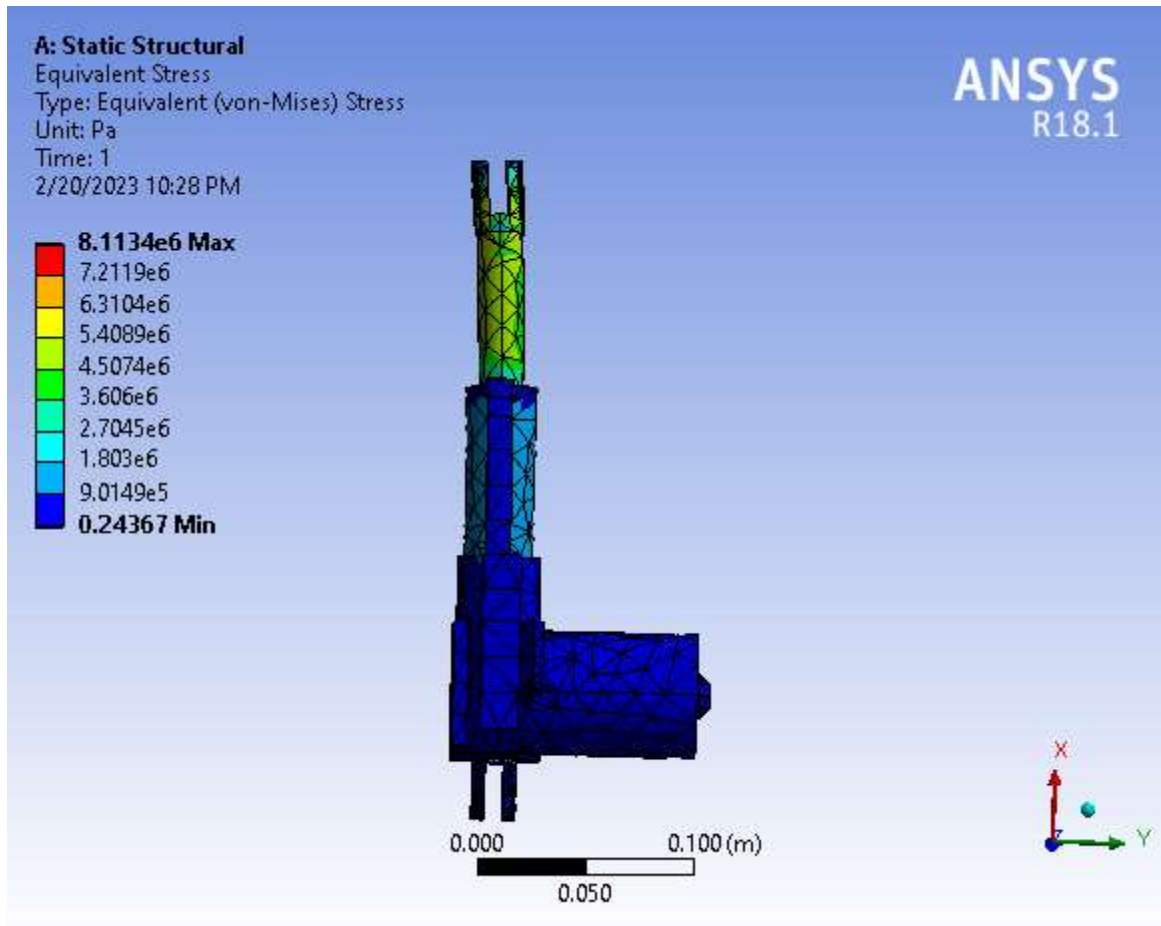


Figure 42: Equivalent (Von-Mises) Stress of Linear Actuator

Table 27: Equivalent (Von-Mises) Stress of Static Frame

	Equivalent (Von-Mises) Stress (Pa)
Minimum	0.24367
Maximum	8.1134×10^6

5.3.2.3 Equivalent Elastic Strain of Linear Actuator

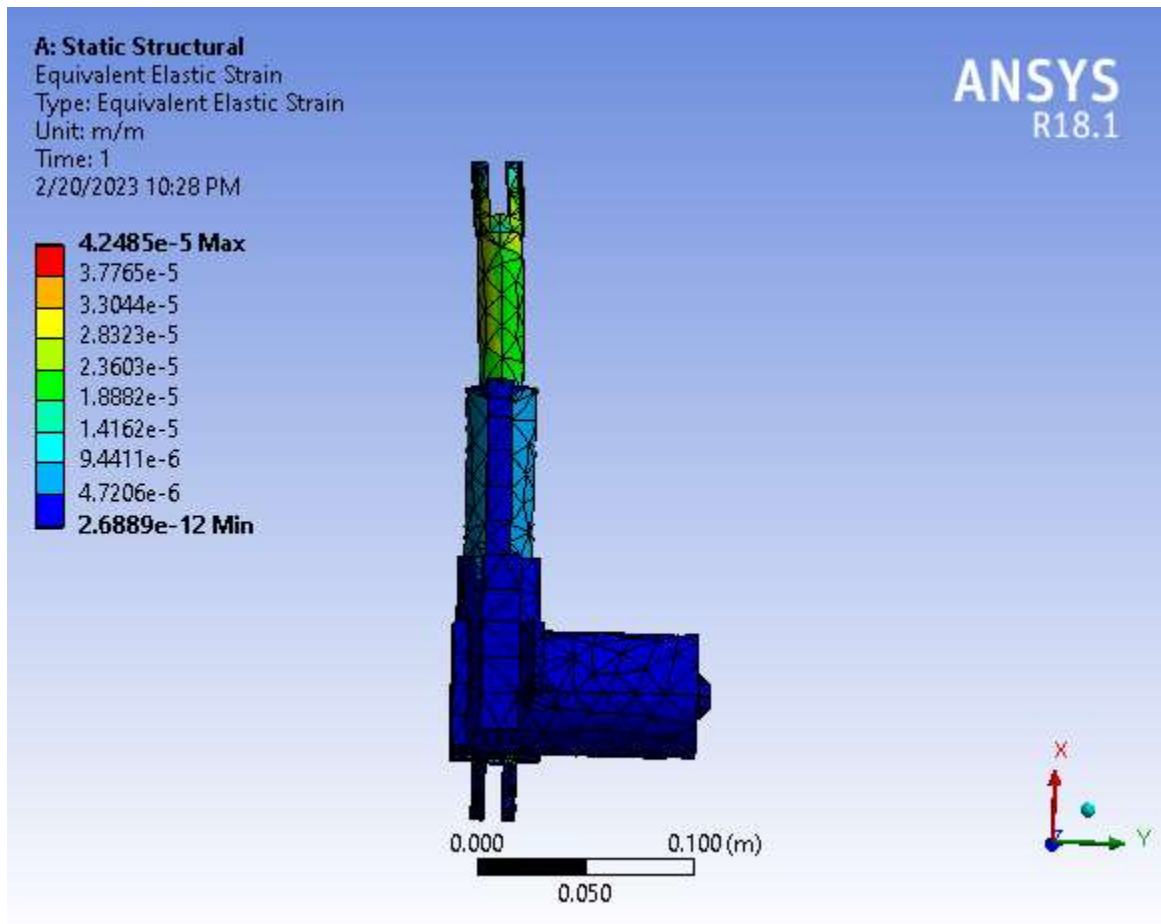


Figure 43: Equivalent Elastic Strain of Linear Actuator

Table 28: Equivalent Elastic Strain of Static Frame

	Equivalent Elastic Strain of Linear Actuator (m/m)
Minimum	2.6889×10^{-12}
Maximum	4.2485×10^{-5}

5.4 Conclusion

We have designed the parts using the software, “SolidWorks” and then determined the total deformation, equivalent (Von-Mises) stress, life cycle and safety factor using “Ansys”. For a structure to be considered safe, its factor of safety must be greater than 1, high factor of safety means a heavier component, more upscale material and an improved design

Chapter 6: Qualitative Analysis of Material, Manufacturing Process and Joining Process Selection

6.1 Introduction

Material selection is one of the most important steps of product design. It is the act of selecting the material that is best suited to meet the requirements of a given application. Various factors determine the selection requirements, such as physical properties, chemical properties, mechanical properties, cost, availability etc. The design of a part depends on the selected material. Improper material selection can impact the required application and safety. Moreover, it is necessary to examine how two or more materials might behave when in contact with each other. With the basic knowledge of material properties and different manufacturing processes available for them, a qualitative analysis of material, manufacturing process and joining process selection has been performed.

6.2 Selection of Materials for Different Sections

Different parts need different materials for manufacturing based on the required properties and expected performance. Materials are chosen depending on various properties such as mechanical, electrical, chemical properties etc. The main objective of material selection is to minimize cost while achieving the desired product performance.

Table 29: Selection of materials for different parts

Section	Parts	Material Selection	Alternative material	Reasons behind Selection of Material
Static Body Frame	Wooden Frame	MDF wood	Plastic, Metal	Durable, Flexibility in manufacturing, Hard to crack

	Drawer Frame	MDF wood	Plastic, Plywood	Durable, Flexibility in manufacturing, Hard to crack
	Cabinet Frame	MDF wood	Plastic, Plywood	Durable, Flexibility in manufacturing, Hard to crack
Accessories	Wheels	Nylon	Rubber, PU	Corrosion resistant, Chemical resistant, Abrasion resistant, Light weight, Easy to clean
Adjustable Body Frame	Control Unit Actuator	Stainless Steel	PSS, PU	High Toughness, High Strength, Corrosion resistant
	Control Unit Wooden Frame	MDF wood	Plastic, Metal	Durable, Flexibility in manufacturing, Hard to crack
	Cabinet Actuator	Stainless Steel	PSS, PU	High Toughness, High Strength, Corrosion resistant
	Cabinet Wooden Frame	MDF wood	Plastic, Metal	Durable, Flexibility in manufacturing, Hard to crack
	Dynamic Base Plate Actuator	Stainless Steel	PSS, PU	High Toughness, High Strength, Corrosion resistant

	Dynamic Base Plate Wooden Frame	MDF wood	Plastic, Metal	Durable, Flexibility in manufacturing, Hard to crack
Joining Part	Nut-Bolt	Stainless Steel	Steel, Aluminum	Corrosion resistant, Good Strength

6.3 Qualitative Analysis of Manufacturing Process Selection

It is crucial to choose the optimum manufacturing process among the ones that apply to any component of the product. We need to invest more time in this in order to lower production costs, metallurgical change, material waste, and increase performance. Cheap, adaptable, and appropriate processes make it simple to repeatedly generate the product. While strict, complicated processes raise the price of the final product significantly. If it is less expensive to outsource the widely accessible standard parts than to produce them, that is an excellent choice.

Table 30: Manufacturing process selection for the parts

Section	Parts	Possible Manufacturing Process	Make/ Outsource	Manufacturing Process Selection	Reason Behind Selection
Static Body Frame	Wooden Main Frame	Carpentry	Make	Carpentry	Available, cheap
	Drawer Frame	Carpentry	Make	Carpentry	Easy to perform, better surface finish

	Drawer Lock	Metal Cutting, Casting, Sheet metal Formation	Outsource		Available, Cheaper
	Cabinet Frame	Carpentry	Make	Carpentry	Available, cheaper
	Cabinet Lock		Outsource		Available, Cheaper
Accessories	Wheel	Metal Cutting, Casting, Milling	Outsource		Available, Cheaper
	Charging Port	Injection Molding	Outsource		Available
	Energy Backup		Outsource		Cost Efficient, Available
	Modular Light		Outsource		Available, Very Cheap
Dynamic Body Frame	Actuator		Outsource		Available
	Control Unit Frame	Carpentry	Make	Carpentry	Easy to perform, better surface finish
	Raspberry Pi Computer		Outsource		Cost Efficient, Available
	Monitor		Outsource		Available

	Sound System		Outsource		Available, Cheaper
	Micro Controller		Outsource		Cost Efficient, Available
	Computer Controlled Cabinet Lock		Outsource		Available, Cheaper
	Wooden Frame	Carpentry	Make	Carpentry	Easy to perform, better surface finish
	Dynamic Base Frame	Carpentry	Make	Carpentry	Available, cheaper

This table was created with availability and price in mind. Some components are simple and less expensive to produce. Some of them are readily available, and outsourcing will save money for them.

6.4 Qualitative Analysis of Parts Joining Method Selection

Joints are basically of three types- permanent, semi-permanent and temporary. Part joining is very crucial for a product development method. So, we have to keep in mind the flexibility of operation, maintenance and modification chances. The joining method selection tables for “Smart Pop-Up Table” is as below:

Table 31: Joint method selection table

Joining Parts	Type of Joint	Joining Method Options	Joint Selected	Reason behind Selection
Wooden frames	Permanent	Glue, Clamping	Glue	Provides sufficient strength to the joint and cost effective.

Frame and Monitor	Temporary	Clamping, Nut-bolt	Clamping	Provides sufficient strength to the joint
Frame and computer	Temporary	Screw clamp, Glue	Screw Clamp	Provides sufficient strength to the joint
Frame and Actuator	Temporary	Clamping, Nut-Bolt	Nut-Bolt	Ease of Motion
Frame and Locking Module	Temporary	Nut-Bolt, Clamp	Nut-Bolt	Provides sufficient strength to the joint
Frame and Power Socket	Temporary	Glue, Nut-Bolt	Glue	Provides sufficient strength to the joint
Frame and Sensor	Temporary	Glue, Clamping	Clamping	Ease of Motion
Frame and Knobs	Temporary	Glue, Nut-Bolt	Nut-Bolt	Provides sufficient strength to the joint
Frame and Wheels	Temporary	Nut-Bolt, Glue, Clamp	Nut-Bolt	Ease of motion and strength

As our product is a portable, easy to operate, modifiable one, so every joint is temporary. Operator can easily explode the design whenever it needs maintenance or modify it.

Table 32:Finishing process selection table

Finishing Parts	Finishing process option	Process selected	Reason behind Selection
Every Wooden Parts	Wax polishing, Vinegar polishing, Spirit Polishing	Wax Polishing	Low time consuming, Cost-effective and safe for electric components

Table 33: Coloring process selection table

Coloring Parts	Coloring process option	Process selected	Reason behind Selection
Every Wooden Parts	Spray Painting, Wax Painting, Dying, Chalk Painting	Chalk Painting	Durable, cost-efficient and safe for electric components

6.5 Conclusion

The qualitative analysis of material, manufacturing process, and joining process selection provides an insight into the overall production process. Qualitative analysis is helpful in a product design process before moving to quantitative analysis. It helps to eliminate the less promising choices for materials, manufacturing process, and joining method very early on and saves us valuable time and computational effort. Qualitative approach of selection may not be meticulous as intuitions have been applied to create it but it certainly leads the way of production. Thus, a more specific goal can be set and more accurate and precise data can be gathered regarding the selection procedures

Chapter 7: Material selection using Weighted Average Method

7.1 Introduction

Material selection is an important part of the product design process. The design of engineering components is limited by the available materials and selection of the wrong material can lead to catastrophic failures. In the previous chapter, a qualitative analysis of material, manufacturing process, and joining method selection has been carried out. But qualitative analysis is nothing but applying intuitive knowledge. Therefore, quantitative analysis for selection criteria is imperative to manufacture a good product.

There are several methods for quantifying the properties of a material or manufacturing process. Here, in this chapter, we are using the “Digital Logic Method” for selecting material based on their performance indices.

7.2 List of Parts for Each Sub-assembly

1. Body Frame
2. Wheels
3. Actuators
4. Computer System

7.3 Equations

1. Relative Emphasis Coefficient, $\alpha = (\text{No. of positive decisions acquired by criteria}) / (\text{Total no. of positive decisions})$

2. For properties to be maximized,

Scaled Property, $\beta = (\text{Numerical value of property} / \text{Maximum value in list}) \times 100$

3. For properties to be minimized,

Scaled Property, $\beta = (\text{Minimum value in list} / \text{Numerical value of property}) \times 100$

4. Weighted score = Relative emphasis coefficient (α) * Scaled property (β)

5. Performance index, $\gamma = \Sigma \text{Weighted score } (\alpha * \beta)$

7.4 Likert Scale

Likert scale is a psychometric scale commonly involved in research that employs questionnaires.

This scale is used for rating different categories according to their significance or intensity.

When

responding to a Likert item, respondents specify their level of agreement or disagreement on a symmetric agree-disagree scale for a series of statements. Thus, the range captures the intensity of

their feelings for the given item.

In this chapter, we are going to use the Likert scale for those criteria that can't be assigned with numerical values. The scale is given below:

Table 34: Likert Scale

Very Good	5
Good	4
Moderate	3
Poor	2
Very Poor	1

7.5 Quantitative Analysis of Material Selection

7.5.1 Material Selection for Body

Table 35: List of numerical values of material property

Property	MDF Wood	Wood	Plastic
Strength (MPa)	31	40-60	10-50
Density(kg/m ³)	720	400-700	900-960
Melting Point(°C)	150-220	150-220	110-130
Hardness (HB)	600-800	150-220	60-90

Table 36: Lists of selection criteria with values from Likert scale

Property	MDF Wood	Wood	Plastic
Cost	5	4	4
Strength	5	4	3
Light Weight	4	3	5
Availability	4	4	4
Aesthetics	3	4	2

Table 37: Determination of relative importance of material selection criteria for Body using Digital Logic Method

Selection Criteria	Number of Positive Decisions, $N=n(n-1)/2=15$										Positive Decisions	Relative Emphasis Coefficient, α
	1	2	3	4	5	6	7	8	9	10		
Cost	0	0	0	1							1	0.1
Strength	1				1	0	1				3	0.3
Light Weight		1			0			0	0		1	0.1
Availability			1			1		1		1	4	0.4
Aesthetics				0			0		1	0	1	0.1
Number of total positive decisions											10	$\Sigma\alpha = 1.00$

Table 38: : Calculation of the performance index for material selection for Body

Selection Criteria	Weighted Factor, α	MDF Wood		Wood		Plastic	
		Scaled Property, β	Weighted Score, $\alpha\beta$	Scaled Property, β	Weighted Score, $\alpha\beta$	Scaled Property, β	Weighted Score, $\alpha\beta$
Cost	0.1	80	8	100	10	100	10

Strength	0.3	100	30	80	24	60	18
Light Weight	0.1	80	8	60	6	100	10
Availability	0.4	100	40	100	40	100	40
Aesthetics	0.1	75	7.5	100	10	50	5
Material Performance Index			93.5		90		83

For the manufacturing of the Body, MDF Wood has the highest weighted average score than Wood and Plastic. So, MDF Wood is selected as the manufacturing material.

7.5.2 Material Selection for Nutbolt

Table 39: List of numerical values of material property

Property	HCS	MS	Nickel Plated Steel
Strength (MPa)	685	550	654
Hardness (Rockwell B)	70	71	60

Table 40: Lists of selection criteria with values from Likert scale

Property	HCS	MS	Nickel Plated Steel
Cost	4	3	5
Strength (MPa)	5	3	4
Hardness	4	5	3
Availability	5	4	3

Table 41: Determination of relative importance of material selection criteria for Nutbolt using Digital Logic Method

Selection Criteria	Number of Positive Decisions, $N = \frac{n(n-1)}{2} = \frac{(4*3)}{2} = 6$						Positive Decisions	Relative Emphasis Coefficient, α
	1	2	3	4	5	6		
Cost	0	0	1				1	0.167
Strength	1			1	1		3	0.5
Hardness		1		0		0	1	0.167
Availability			0		0	1	1	0.167
Total Number of Positive Decision							6	$\Sigma\alpha = 1.00$

Table 42: Calculation of the performance index for material selection for Nutbolt

Selection Criteria	Weighted Factor, α	HCS		MS		Nickel Plated Steel	
		Scaled Property, β	Weighted Score, $\alpha\beta$	Scaled Property, β	Weighted Score, $\alpha\beta$	Scaled Property, β	Weighted Score, $\alpha\beta$
Cost	0.167	75	12.525	100	16.7	60	10.02
Strength	0.5	100	50	60	30	80	40
Hardness	0.167	80	13.36	100	16.7	60	10.02
Availability	0.167	100	16.7	80	13.36	60	10.02
Material Performance Index			92.585		76.76		70.06

Material performance index of High Carbon Steel (HCS) is largest in comparison of the rest values. So, High Carbon Steel is selected for Nut-Bolt.

7.5.3 Material Selection for Glue/Adhesive

Table 43: List of numerical values of material property

Property	Epoxy	PVA (polyvinyl acetate) glue	Titebond
Strength	685	550	654

Water Resistance	70	71	60
------------------	----	----	----

Table 44: Lists of selection criteria with values from Likert scale

Property	Epoxy	PVA (polyvinyl acetate) glue	Titebond
Cost	5	3	4
Strength	5	3	4
Water Resistance	5	4	3
Availability	3	5	4

Table 45: Determination of relative importance of material selection criteria for Glue/Adhesive using Digital Logic Method

Selection Criteria	Number of Positive Decisions, $N = \frac{n(n-1)}{2} = \frac{(4*3)}{2} = 6$						Positive Decisions	Relative Emphasis Coefficient, α
	1	2	3	4	5	6		
Cost	0	0	1				1	0.167
Strength	1			1	0		2	0.333
Water Resistance		1		0		0	1	0.167
Availability			0		1	1	2	0.333
Total Number of Positive Decision							6	$\Sigma\alpha = 1.00$

Table 46: Calculation of the performance index for material selection for Glue/Adhesive

Selection Criteria	Weighted Factor, α	Epoxy		PVA (polyvinyl acetate) glue		Titebond	
		Scaled Property, β	Weighted Score, $\alpha\beta$	Scaled Property, β	Weighted Score, $\alpha\beta$	Scaled Property, β	Weighted Score, $\alpha\beta$
Cost	0.167	100	16.7	60	10.02	80	13.36
Strength	0.333	60	19.8	100	33.3	75	24.97
Water Resistance	0.167	60	10.02	75	12.52	100	16.7

Availability	0.333	100	33.3	60	19.8	75	24.975
Material Performance Index			79.82		75.64		80.005

Material performance index of Titebond is largest in comparison of the rest values. So, Titebond is selected as adhesive.

7.5.4 Material Selection for Clamp

Table 47: List of numerical values of material properties

Property	Wood	Stainless Steel
Compressive Strength (MPa)	52	310
Density (g/cc)	1.5	8
Modulus of Elasticity (GPa)	14	193

Table 48: Lists of selection criteria with numerical values or values from Likert scale for material selection of clamp

Selection Criteria	Wood	Stainless Steel
Cost	2	5
Surface Finish	3	4

Table 49: Chart for Calculating Relative Emphasis Co-efficient.

Selection Criteria	Number of positive decisions, $N=n(n-1)/2=5(5-1)/2=10$										Positive decisions	Relative emphasis Co-efficient, α
	1	2	3	4	5	6	7	8	9	10		
Cost	1	0	1	1							3	0.3
Surface Finish	0				0	0	1				1	0.1

Compressive Strength (MPa)		1			1			1	1		4	0.4
Density (g/cc)			0			1		0		0	1	0.1
Modulus of Elasticity (GPa)				0			0		0	1	1	0.1
Total Number of Positive Decisions											10	$\sum \alpha = 1.00$

Table 50: Chart for Calculating Performance Index

Selection Criteria	Weighting Factor, α	Wood		Stainless Steel	
		Scaled Property, β	Weighted Score, $\alpha\beta$	Scaled Property, β	Weighted Score, $\alpha\beta$
Cost	0.3	100	30	40	12
Surface Finish	0.1	75	7.5	100	10
Compressive Strength (MPa)	0.4	16.77	6.71	100	40
Density (g/cc)	0.1	100	10	18.8	1.88
Modulus of Elasticity (GPa)	0.1	7.25	.725	100	10
Material Performance Index, γ			54.94		73.88

Result: Process performance index is greater (73.88) for stainless steel compared to clamp. Thus, stainless steel is selected as the preferred material for the clamps.

7.6 Quantitative analysis for joining process selection

7.6.1 Joining Process selection for Wooden Frame

Table 51: Lists of selection criteria with numerical values or values from Likert scale for joining process of wooden frame

Selection Criteria	Adhesive	Nails	Nut-Bolt
Cost	1	2	3
Strength	5	3	4
Surface Finish	5	4	3

Table 52: Chart for Calculating Relative Emphasis Co-efficient.

Selection Criteria	Number of positive decisions, $N=n(n-1)/2=3(3-1)/2=3$			Positive decisions	Relative emphasis Co-efficient, α
	1	2	3		
Cost	0	1		1	0.333
Strength	1		0	1	0.333
Surface Finish		0	1	1	0.333
Total Number of Positive Decisions				3	$\sum \alpha = 1.00$

Table 53: Chart for Calculating Performance Index

Selection Criteria	Weighting Factor, α	Adhesive		Nails		Nut-Bolt	
		Scaled Property, β	Weighted Score, $\alpha\beta$	Scaled Property, β	Weighted Score, $\alpha\beta$	Scaled Property, β	Weighted Score, $\alpha\beta$
Cost	0.333	100	33.3	50	16.65	33.33	11.1
Strength	0.333	100	33.3	60	19.98	80	26.64

Surface Finish	0.333	100	33.3	80	26.64	60	19.98
Material Performance Index, γ			99.9		63.27		57.72

Result: Process performance index is greater (99.9) for Adhesive compared to other alternatives. Thus, adhesive is selected as the joining process for the wooden frame.

7.6.2 Joining Process selection for Actuator

Table 54: Lists of selection criteria with numerical values or values from Likert scale for joining process of actuator

Selection Criteria	Clamp	Nut-Bolt
Cost	3	2
Strength	4	3
Availability	3	4
Design Flexibility	3	4

Table 55: Chart for Calculating Relative Emphasis Co-efficient.

Selection Criteria	Number of positive decisions, $N=n(n-1)/2=4(4-1)/2=6$						Positive decisions	Relative emphasis Co-efficient, α
	1	2	3	4	5	6		
Cost	0	1	1				2	.333
Strength	1			1	0		2	.333

Availability		0		0		1	1	.167
Design Flexibility			0		1	0	1	.167
Total Number of Positive Decisions							6	$\sum \alpha = 1.00$

Table 56: Chart for Calculating Performance Index

Selection Criteria	Weighting Factor, α	Clamp		Nut-Bolt	
		Scaled Property, β	Weighted Score, $\alpha\beta$	Scaled Property, β	Weighted Score, $\alpha\beta$
Cost	.333	66.67	22.2	100	33.3
Strength	.333	100	33.3	75	24.98
Availability	.167	75	12.53	100	16.7
Design Flexibility	.167	75	12.53	100	16.7
Material Performance Index, γ			50.56		91.68

Result: Process performance index is greater (91.68) for Nut-Bolts compared to clamp. Thus, Nut-Bolt is selected as the joining process for the wooden frame.

7.6.3 Joining Process selection for Computer System

Table 57: Lists of selection criteria with numerical values or values from Likert scale for joining process of computer system

Selection Criteria	Clamp	Nut-Bolt
Cost	3	2
Strength	4	3
Availability	3	4
Design Flexibility	3	4
Versatility	4	2

Table 58: : Chart for Calculating Relative Emphasis Co-efficient.

Selection Criteria	Number of positive decisions, $N=n(n-1)/2=5(5-1)/2=10$										Positive decisions	Relative emphasis Co-efficient, α
	1	2	3	4	5	6	7	8	9	10		
Cost	0	1	0	0							1	0.1
Strength	1				1	1	1				4	0.4
Availability		0			0			1	0		1	0.1
Design Flexibility			1			0		0		0	1	0.1
Versatility				1			0		1	1	3	0.3
Total Number of Positive Decisions											10	$\sum \alpha = 1.00$

Table 59: Chart for Calculating Performance Index

Selection Criteria	Weighting Factor, α	Clamp		Nut-Bolt	
		Scaled Property, β	Weighted Score, $\alpha\beta$	Scaled Property, β	Weighted Score, $\alpha\beta$
Cost	0.1	66.67	6.67	100	10
Strength	0.4	100	40	75	30
Availability	0.1	75	7.5	100	10
Design Flexibility	0.1	75	7.5	100	10
Versatility	0.3	100	30	50	15
Material Performance Index, γ			91.67		75

Result: Process performance index is greater (91.67) for clamps. Thus, clamp is selected as the joining process for the wooden frame.

7.7 Quantitative analysis for finishing process selection

7.7.1 Selection of finishing process selection for Smart Pop-Up Table

Table 60: Lists of selection criteria with numerical values or values from Likert scale for Wax Polishing and Burnishing

Selection criteria	Wax Polishing	Burnishing
Cost	4	5
Smoothness	5	4
Availability	4	4
Material wastage	5	3
Durability	4	3

Table 61: Determination of Relative Importance of Goals Using Digital Logic Method

Selection criteria	Number of positive decisions, $N=n(n-1)/2=5(5-1)/2=10$										Positive Decision	Relative Emphasis Co-efficient, α
	1	2	3	4	5	6	7	8	9	10		
Cost	0	0	1	0							1	0.10
Smoothness	1				0	0	0				1	0.10
Availability		1			1			1	0		3	0.30
Material wastage			0			1		0		0	1	0.10
Durability				1			1		1	1	4	0.40
Total Number of Positive Decision											10	$\Sigma\alpha = 1.00$

Table 62: Calculation of performance index for selecting finishing process

Selection criteria	Weighting Factor, α	Wax Polishing		Burnishing	
		Scaled Property, β	Weighted Score, $\alpha\beta$	Scaled Property, β	Weighted Score, $\alpha\beta$
Cost	0.10	100	10	80	8
Smoothness	0.10	100	10	80	8
Availability	0.30	100	30	100	30
Material wastage	0.10	60	6	100	10
Durability	0.40	100	40	75	30
Material Performance Index, γ		96		86	

From the table, it is found that Wax Polishing has the most relative score over Burnishing. So, Wax Polishing is selected as the mode of finishing process of the ‘Smart Pop-Up Table’.

7.8 Quantitative analysis for coloring process selection

7.8.1 Selection of coloring process selection for Smart Pop-Up Table

Table 63: Lists of selection criteria with numerical values or values from Likert scale for painting with liquid color and color spray

Selection criteria	Chalking	Color spray
Cost	5	4
Durability	5	4
Availability	4	5
Wastage of color	5	4
Smoothness	5	3

Table 64: Determination of Relative Importance of Goals Using Digital Logic Method

Selection criteria	Number of positive decisions, $N=n(n-1)/2=5(5-1)/2=10$										Positive Decision	Relative Emphasis Co-efficient, α
	1	2	3	4	5	6	7	8	9	10		
Cost	0	0	0	1							1	0.10
Durability	1				0	1	1				3	0.30
Availability		1			1			1	0		3	0.30
Wastage of color			1			0		0		0	1	0.10
Smoothness				0			0		1	1	2	0.20
Total Number of Positive Decision											10	$\Sigma\alpha = 1.00$

Table 65: Calculation of performance index for selecting coloring process

Selection criteria	Weighting Factor, α	Chalking		Color spray	
		Scaled Property, β	Weighted Score, $\alpha\beta$	Scaled Property, β	Weighted Score, $\alpha\beta$

Cost	0.10	80	8	80	8
Durability	0.30	100	30	80	24
Availability	0.30	80	24	100	30
Wastage of color	0.10	80	8	100	10
Smoothness	0.20	100	20	60	12
Material Performance Index, γ	90			84	

From the table, it is found that Chalking has the most relative score over Color spray. So, Chalking is selected as the mode of finishing process of the ‘Smart Pop-Up Table’.

Chapter 8: Cost & Sensitivity Analysis

8.1 Introduction

Cost is inextricably linked to a product's design because the cost is heavily influenced by design decisions. The major elements for determining a product's economic aspect are cost and income. The goal is to cut costs and always increase profits. Cost analysis demonstrates a product's ability to produce a desired level of output while making a profit. It demonstrates a product's ability to achieve its break-even point.

8.2 Cost Analysis.

8.2.1 Market Analysis:

In Bangladesh, there is a growing demand for innovative and efficient solutions for studies and work. The market for this product is students and professionals who are looking for a space-saving and secure solution. With the increasing trend of digitalization in Bangladesh, there is a potential for this product to appeal to a large segment of the market.

8.2.2 Financial Projections:

The cost of production for each unit of the Smart Study Table is around 35000 tk. The expected retail price for each unit is 45000 tk. Based on market research and projections, it is estimated that the company can sell around 750 units in the first year and 1200 units in the second year. The estimated revenue for the first year is 33,750,000 taka.

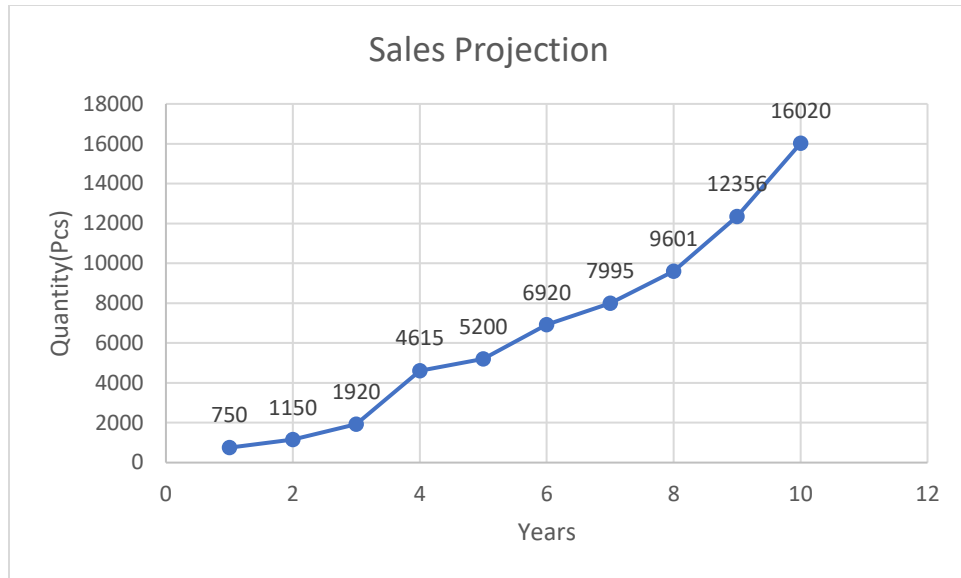


Figure 44: Sales Projection

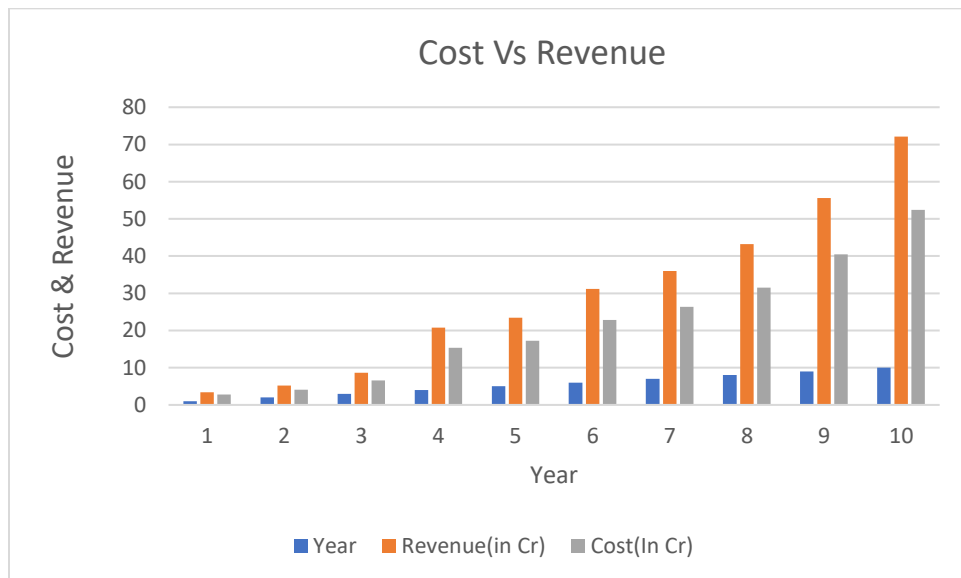


Figure 45: Cost vs Revenue Stream

8.2.3 Fixed costs:

To calculate the yearly fixed cost, we need to sum up all the fixed costs for one year.

Production Manager Salary: $30000 \times 12 = 360000$

Quality Control Manager Wage: $25000 \times 12 = 300000$

Product Development Officer Salary: $30000 \times 12 = 360000$

Security Guard Salary: $10000 \times 12 = 120000$

Clarke Salary: $12000 \times 12 = 144000$

Marketing Manager Salary: $20000 \times 12 = 240000$

Bank Loan Interest: $2000000 \times 15/100 = 300000$

Insurance: 30000 tk

Transportation: 500000 tk

Yearly Rent: 500,000 tk

Monthly Utility: $20000 \times 12 = 240000$

Advertising and Marketing: $50000 \times 12 = 600000$

Summing up all the fixed costs,

Yearly Fixed Cost = $360000 + 300000 + 360000 + 120000 + 144000 + 240000 + 300000 + 30000$
 $+ 500000 + 240000 + 600000 + 500000$

Yearly Fixed Cost = 3,69,4000 tk

So, the yearly fixed cost is 3,69,4000 tk.

8.2.4 Variable costs:

Monitor 3050

VGA to HDMI Cable 150

Electronics 1715

5 V charger 250

Table 5500

Actuators 12500

220 V AC to 12V DC converter 160

Screws 100

Orange Pi 3780

Adata SD Card 450

Key Board 370

2 wire USB 50

Wood Booshing 100

Drawer Channel 220

2 pc Drawer Channel 420

30 ft Wire	250
Wall pin	60
Socket	50
Adapters	800
switch+socket+Board	220
Smart Switch	650
Wi-Fi Receiver	350
2 to 3 pin converter	70
plug(2Pc)	40

Labor costs:

6 labor hours per study table x 200 taka/day per employee = 1200 taka

Total variable cost per study table = 31275
+ 32475 = 32475 taka

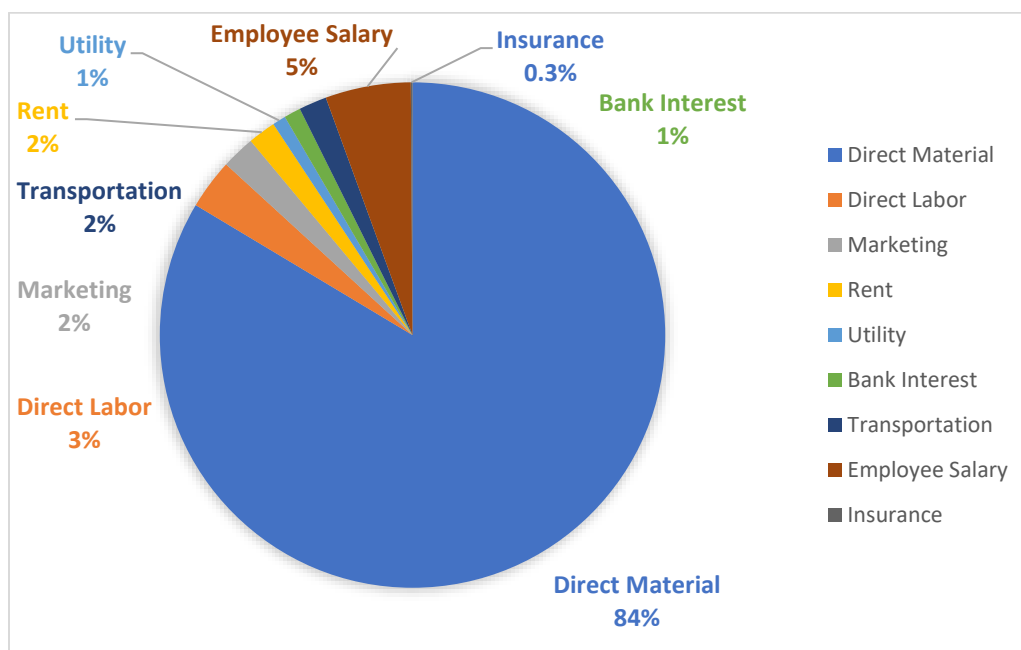


Figure 46: Investment Breakdown

8.2.5 Break Even Analysis

Fixed Cost = 3,69,4000 tk

Fixed cost per unit = 3,69,4000 /750 tk (Demand expected to be 750 units per year) = 4926 tk

The formula for break-even point is:

$BEP = \text{Total Fixed Cost} / (\text{Selling Price} - \text{Variable Cost per Product})$

Plugging in the values:

$BEP = 3700000 / (45000 - 32475)$

BEP = 296 products

So the break-even point is 347 products, rounded off to the nearest whole number. This means if the company produces and sells more than 347 products, it will start to make a profit.

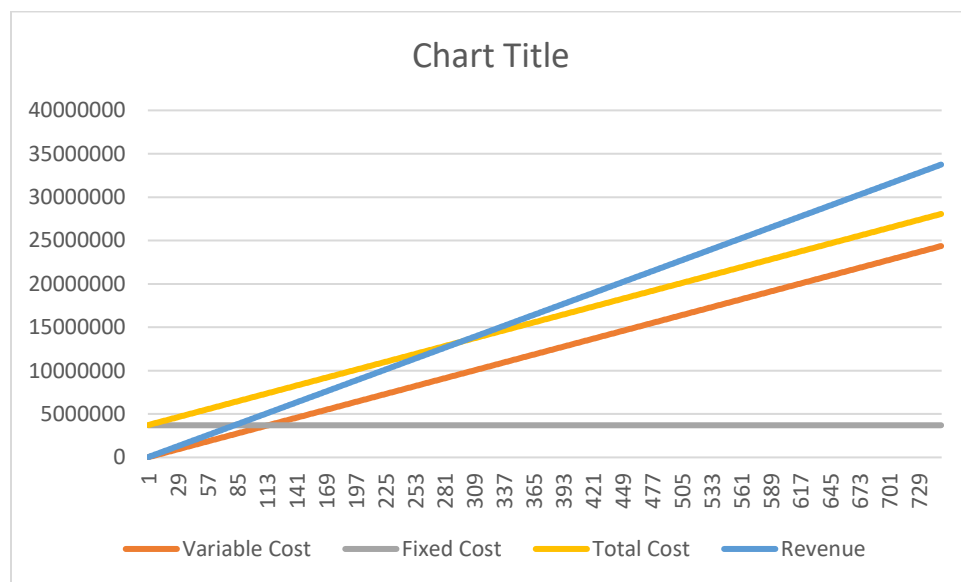


Figure 47: Break Even Analysis

Break-even point will be reached within 6 months.

8.3 Sensitivity Analysis

Sensitivity analysis gives an insight about how sensitive the break-even point is with respect to a particular cost or revenue parameter. In this sensitivity analysis, a change of 10% is made for each cost parameter separately and then break-even point is computed. This analysis will provide the list of sensitive parameters.

8.3.1 Sensitivity of Demand

Demand per year 750 unit

After increasing demand 10%, new demand = $750 + 750 \times 0.1$
= 825 unit

Variable cost per unit = 32475/-

At Break Even Point,

Selling price $\times$ Qbep = Total Fixed cost + Variable cost $\times$ Qbep

Or, $45000 \times \text{Qbep} = 3700000 + (32475 \times \text{Qbep})$

Or, Qbep = 296 units.

$\Delta \text{Qbep} = 0\%$

So, the demand has no effect on breakeven point. The breakeven point is not sensitive to demand.

8.3.2 Sensitivity of Direct Material

Total cost of direct material = 24356250/-

After 10% increase, total cost of direct material = 24356250×1.10
= 26791875/-

Total variable cost will be = $900000 + 26791875$
= 27691875/-

Variable cost per unit = $27691875 / 750$
= 36922/-

At Break Even Point,

Selling price $\times$ Qbep = Total Fixed cost + Variable cost $\times$ Qbep

Or, $45000 \times \text{Qbep} = 3700000 + (36922 \times \text{Qbep})$

Or, Qbep = 458 units

$$\Delta Q_{bep} = (458 - 296)/296 * 100\%$$

$$= 54.73\%$$

So, the breakeven is too much sensitive to direct material cost.

8.3.3 Sensitivity of Direct Labor

Total cost of labor per Year = 900000/-

After 10% increase, total cost of labor = 900000*1.10
= 990000/-

Total variable cost will be = 26791875 /-+ 990000
= 27781875/-

Variable cost per unit = 26791875/750
= 37042.5/-

At Break Even Point,

Selling price $\times$ Q_{bep} = Total Fixed cost+ Variable cost $\times$ Q_{bep}

Or, 45000 $\times$ Q_{bep} = 3700000+ (37042.5 $\times$ Q_{bep})

Or, Q_{bep} = 465 units.

$$\Delta Q_{bep} = (465 - 296)/296 * 100\%$$

$$= 67.23\%$$

So, the direct material is sensitive to the breakeven point. It can increase the breakeven enormously.

8.3.4 Sensitivity of Fixed Cost

Total fixed cost= 3700000/-

After 10% increase, the fixed cost= 3700000*1.10
= 4073400/-

At Break Even Point,

Selling price $\times$ Q_{bep} = Total Fixed cost+ Variable cost $\times$ Q_{bep}

Or, 45000 $\times$ Q_{bep} = 4073400+ (32475 $\times$ Q_{bep})

Or, Q_{bep} = 325 units.

$$\Delta Q_{bep} = (325 - 296)/296 * 100\%$$

$$= 10\%$$

So, the direct material is sensitive to the breakeven point. The cost of direct material can increase the breakeven point drastically.

The cost analysis is concerned with determining the monetary value of inputs (labor, raw material), called the overall cost of production which helps in deciding the optimum level of production. Cost analysis helped us estimate the payback period by examining the break-even point

8.4 Conclusion

Product design is a broad term that refers to the efficient generation and development of ideas into new goods through a process. As a result, it's a crucial part of new product development. Product design can be a simple fix for a fault that has to be fixed, or it can be a brand-new concept with the potential to thrive. As there is no current producer of our product, it will attract customers' attention to buy it. It will save money and space as it can serve multiple purposes. We aim to design and produce a new product to fulfill the requirements of multiple products in one product. Studying product design gives us broad theoretical knowledge and the practical skillset that we need to create new products.

Glossary:

- **Active sensors:** Sensors that detect and respond to movement or changes in the environment.
- **Ansys:** A software used for engineering simulation and design analysis.
- **Arduino Uno:** A microcontroller board used in the Smart Pop-Up Table.
- **Auto adjustment module:** A module that allows a product to self-adjust to meet changing needs or circumstances.
- **Automated sieving machine:** A machine used for filtering materials, attached to a movable frame with bearings and powered by an electric motor connected through a belt drive mechanism.
- **Black Box Analysis:** A tool used during functional decomposition that considers the product as a black box with an unknown mechanism.
- **Black Box Model:** An abstract model of a product that shows inputs and outputs to describe a technical system or product.
- **Break-even point:** the level of sales at which the revenue generated is equal to the cost of production.
- **Central computer:** A computer system that controls a product's functioning.
- **Chalking:** A coloring process that involves applying a powdery substance to a surface to create a chalk-like appearance.
- **Cluster Function:** A representation of the flow of energy and the main working principle of the product.
- **Clutter -** A collection of things lying about in an untidy and disorganized state.
- **Component Hierarchy:** A classification of a product into its essential components to determine what components are required for the product's design.
- **Customer-driven processes:** Processes that are designed with customer needs in mind.
- **Digital logic method:** A method of determining relative emphasis coefficients that involve assigning numerical values to positive decisions for each selection criterion.
- **Engineering Specifications:** Technical details that define how a product will function.
- **Equivalent elastic strain:** The amount of strain in a material that is caused by a particular stress.

- Ergonomic: A design that considers the user's comfort and efficiency, accommodating their position and posture.
- Factor of safety: The ratio of the maximum stress a material can withstand to the applied stress.
- FAST Diagram: A diagram that represents the flow of data, materials, and information through the product.
- Function Tree: A hierarchical representation of the functions required to achieve the prime function.
- Functional Decomposition: A technique for breaking down a larger task into smaller tasks to better understand complex functionalities.
- House of Quality: A visual tool used to map out the relationship between customer requirements and technical specifications.
- Hue: The color or shade of a surface or material.
- Iterative: The process of repeating a task or series of tasks until the desired outcome is achieved.
- Life cycle: The stages a product goes through from initial design to disposal.
- Likert scale: A scale used to rate or measure attitudes or opinions, typically ranging from strongly agree to strongly disagree.
- Linear Actuator: A device that converts rotational motion into linear motion.
- Linearly actuating mechanism: A mechanism that moves in a straight line.
- Logical Decomposition: A design approach that involves breaking down a larger task into smaller sub-tasks in a logical manner.
- Malaysian MDF wood: Medium-density fiberboard made from Malaysian timber.
- Material performance index: A measure of a material's performance relative to other options.
- Metallurgical change: Changes to the microstructure of a material due to exposure to heat, pressure, or other conditions during manufacturing. Metallurgical changes can affect the mechanical and chemical properties of a material..

- **Outsource:** To contract with an outside company or individual to perform a task or provide a service that is normally done in-house. Outsourcing can be used to reduce costs, increase efficiency, or access specialized expertise.
- **Performance criteria:** Standards used to measure the performance of a product.
- **Performance index:** A measure of how well a process or material performs relative to other options.
- **Physical prototyping:** The process of creating a physical model of the product being developed.
- **Pigment:** A substance that is used to add color to a material or surface.
- **Pop-up mechanism:** A mechanism that allows a component of a product to be raised or lowered.
- **Pop-up screen -** A screen or display that appears when needed and disappears when not in use, typically hidden within a device or object.
- **Preventive maintenance:** Maintenance performed on a product to prevent future problems.
- **Prototype:** A preliminary version of a product used for testing and evaluation.
- **Qualitative analysis:** A non-quantitative approach to data analysis that focuses on understanding and interpreting data based on subjective observations and opinions.
- **Quality Function Deployment (QFD):** A technique used to convert customer needs into technical specifications for a product.
- **Quantitative analysis:** An analysis that uses numerical data to make a decision or draw a conclusion.
- **Raspberry Pi Computer:** A single-board computer designed for teaching computer science and coding, equipped with a microprocessor, memory, and input/output ports.
- **Recursion:** The process of applying functional decomposition recursively until the design team can readily envision ways of achieving each of the sub-tasks.
- **Refinement:** The process of ensuring that all functions required for the product are included in the functional decomposition diagram.
- **Relative emphasis coefficient:** A numerical value assigned to each selection criterion based on its importance or priority.

- Safety factor: A measure of the product's ability to withstand stresses beyond the expected load.
- Scaled property: A value that is scaled to represent the relative importance or effectiveness of a selection criterion.
- Sensitivity analysis: an analysis of the effect of changes in variables on the profitability of a product
- Sensitivity of total revenue: the effect of changes in total revenue on the profitability of a product
- Service life: The duration for which a product or service can be used effectively.
- Static structural analysis: The analysis of how a structure responds to a static load.
- Stress analysis: The analysis of how a structure or material responds to external forces.
- Sub-Assembly: A smaller assembly of components within the product.
- Sub-Function: A smaller task that must be completed to achieve a larger function during the functional decomposition process.
- Versatility: The ability of a process or material to be used in a variety of different applications or settings.
- Voice of Customer: The customers' opinions, needs, and demands regarding a product or service.
- Von-Mise's stress: A measure of the maximum equivalent stress in a structure.
- Wax polishing: A finishing process that involves applying wax to a surface to create a smooth and shiny finish.
- Weighting factor: A numerical value assigned to each customer requirement to reflect its level of importance.

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